

REVISITING DOMESTIC FINANCIAL MARKETS, ASSET MARKETS & FINANCIAL INDICATORS

– Mridul Saggar

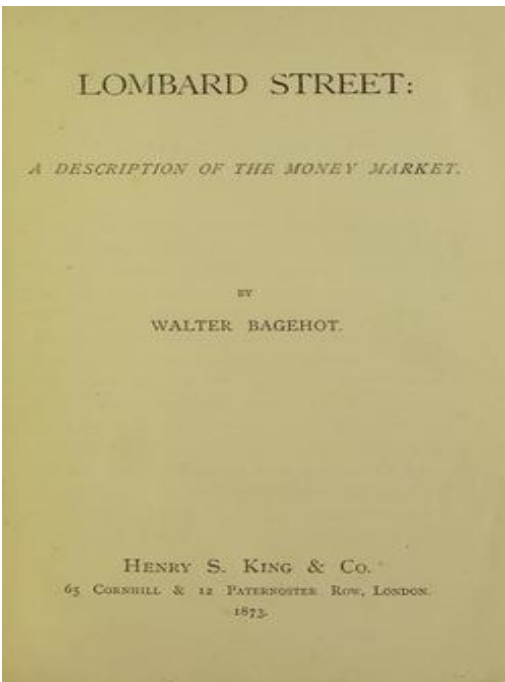
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Lombard Street and the Money Markets in History

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- **Walter Bagehot (1826-1877)** was one of the first writers to describe and explain the money markets to the world of banking and finance in this book which is a treatise. The book was initially printed in Great Britain by Henry S. King & Co. in 1873.
- Bagehot in his book argued that money markets were something real and have got developed as a sophisticated loan-fund in Lombard Street, acting as a crucial intermediary between savings in agricultural areas and the industrial demands of growing urban centers, promoting a functioning economy reliant on borrowed capital.
- The book was written as a reaction to the financial collapse of Overend, Gurney & Co., a wholesale discount bank located at 65 Lombard Street - a historic center of the London banking industry from medieval times to the 1980s. When Gurney & Co suspended payments on 10 May 1866, panic spread across London, Liverpool, Manchester, Norwich, Derby and Bristol
- **BAGEHOT RULE:** To avert a financial crisis driven by panic, the central bank should lend early and liberally to solvent firms facing liquidity problems against good collateral at high rates

What is Money Market?

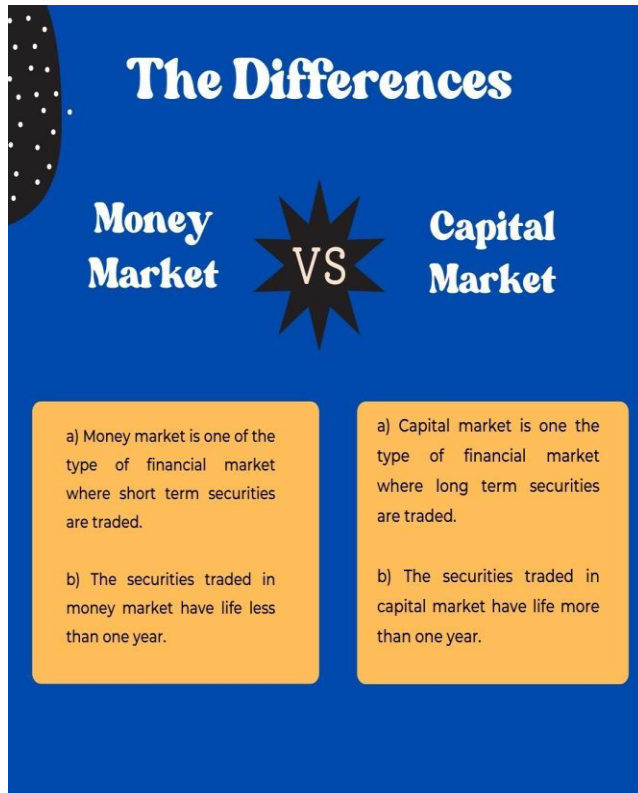
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- **Money Market** is today clearly defined as a marketplace where banks or other financial intermediaries exchange short-term funds or financial instruments for a period of a year or less.
- In India, the concept of money market funds can be traced to pre-Independence era but the market was largely unorganized and segmented in those days
- In 1985 when the money market settlement fund was established to diversify stock investments and enhance liquidity in the financial system.
- In 1987, the RBI Report of the Working Group on the Money Market (Chairman: **N. Vaghul**) laid down a roadmap for development of the money market in India and made recommendations for improving functioning of call money market, developing bills rediscounting market, introducing short-term money market operations in government paper, developing other money market instruments like inter-bank participation certificates & CDs and setting up a finance house to discount papers.
- Since that point money markets in India have evolved, grown in size and become important part of the integrated financial markets here volatilities spillover.

Money Markets in India...(1): Uncollateralized Segment

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- **Call Money Market** is a marketplace where banks exchange funds uncollateralized overnight. (i) Scheduled Commercial Banks (SCBs) including (Regional Rural Banks (RRBs) but excluding Local Area Banks (LABs), (ii) Small Finance Banks (SFBs), (iii) State Co-operative Banks (StCBs), District Central Cooperative Banks (DCCBs) & Urban Cooperative Banks (UCBs) and (iv) Primary Dealers (PDs) can participate in the call, notice & term markets both as lenders & borrowers.
- **Notice Money Market:** When unsecured money is borrowed or lend by eligible banks & PDs for period between 2 days and 14 days it is known as 'Notice Money'.
- **Term Money Market:** When unsecured money is borrowed or lend by eligible banks & PDs for period exceeding 14-days and up to 1-year it is known as 'Term Money Market'.

Money Markets in India...(2): The Repo Market

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- **Repurchase Agreements (Repos):** A repo is a contractual arrangement between two parties, where one party agrees to sell securities to another party at a specified price with a commitment to buy the securities back at a later date for another (usually higher) specified price. A mirror image of this transaction will be the Reverse Repo, where one party agrees to buy the security from another counterparty with a commitment to reverse it after a day or a pre-agreed time period. If the transactions are done without a commitment to reverse the deal, they become outright transactions and they do not fall under repo markets.
- Repos are conducted by central bank with commercial banks (**central bank repo**) or by two eligible counterparties directly (**market repo**). Such transactions for tenors up to 1-year is classified as a money market instrument. Longer-term repos that are not part of money market can also be done. However, the overnight segment is dominant in the repo market.
- At present, repoable securities include Central Government dated securities (G-Secs), Treasury Bills (T-Bills), State Development Loans (SDLs) and Corporate Bonds. Recently, RBI also permitted re-repo of securities acquired under reverse repo subject to certain restrictions. Such an arrangement is permitted only in case of dated G-Secs, T-Bills and SDLs.

Money Markets in India...(3): The CROMS in the Repo Market

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- The entities permitted to undertake repo transactions include Scheduled Commercial Banks, Co-operative Banks, Primary Dealers, Mutual Funds, Insurance Companies and corporate entities. These authorized entities use CROMS to conduct both **Basket Repos** (where a member can borrow against any one of the security forming part of the basket) and **Special Repos** (where a member borrows against specific Central or State Government security rather than a general pool of securities) and to execute Government Securities Lending Transactions (**GSLT**) where an entity which is Gilt Account Holders (GAH) temporarily borrows one specific government security against another security for a fee.
- Clearcorp Repo Order Matching System (**CROMS**) is an STP enabled anonymous **Order Matching Platform** launched by Clearcorp Dealing Systems (India) Ltd, a wholly-owned subsidiary of CCIL, for facilitating dealing in Market Repos in all kinds of Government Securities. OTC transactions (where counterparties are known) too are permitted on this platform & reported. However, market has migrated substantially to Order Matching & OTC trades on Clearcorp or old style outside Clearcorp are now few.

Money Markets in India...(4): Tripartite Repos (TREPS)

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- **Collateralized Borrowing & Lending Obligation (CBLO):** CBLO **was** a money market instrument introduced with effect from January 20, 2003 by the Clearing Corporation of India (CCIL). CCIL was set up with support of the RBI in April 2001 and since 2014 has been acting as a as a Qualified Central Counterparty (QCCP) for RBI regulated markets. A CCP is a financial institution that takes on counterparty credit risk between parties to a transaction. CCIL provides clearing and settlement services for trades in money, G-sec & FX markets.
- **TREPS (Tri-party repo):** is a repo contract where a third entity (apart from the borrower and lender), called a Tri-Party Agent, acts as an intermediary between the two parties to the repo to facilitate services like collateral selection, payment and settlement, custody and management during the life of the transaction. CCIL introduced TREPS from November 3, 2018 under RBI guidelines replacing CBLO. TREPS is very much like CBLO with anonymity of the counterparties assured in both cases but in TREPS. However, TREPS amounted to legal recognition of CCIL as a third repo agent.
- **Eligible entities for TREPS** transactions are Scheduled Commercial banks (SCBs) including Payment Banks & Small Finance Banks, Financial Institutions, Insurance Companies, Mutual Funds (MFs), Primary Dealers (PDs) – standalone or bank PDs, NBFCs, Corporates including SEs & their clearing corps, Provident/ Pension Funds, are participants. These are wide set of repo eligible entities. Retail/ individual investors are prohibited from direct participation.
- CCIL operates the **F-TRAC platform**, which is the designated **trade repository for reporting secondary market trades in corporate bonds, CPs & CDs.**

DEADLINE EXTENDED BY 18 MONTHS

➤ On October 31, 2022, regulator European Securities and Markets Authority (ESMA) withdrew recognition of six Indian central clearing parties (CCPs), including Clearing Corporation of India (CCIL)	➤ ESMA required CCPs to sign cooperation agreements with European regulators. But the RBI refused permission to CCPs to submit to an external regulatory auditor like ESMA
➤ A CCP serves as an intermediary in securities transactions between buyers and sellers	➤ ESMA now has extended the deadline to withdraw membership from CCIL to October 2024

Money Markets in India... (5):

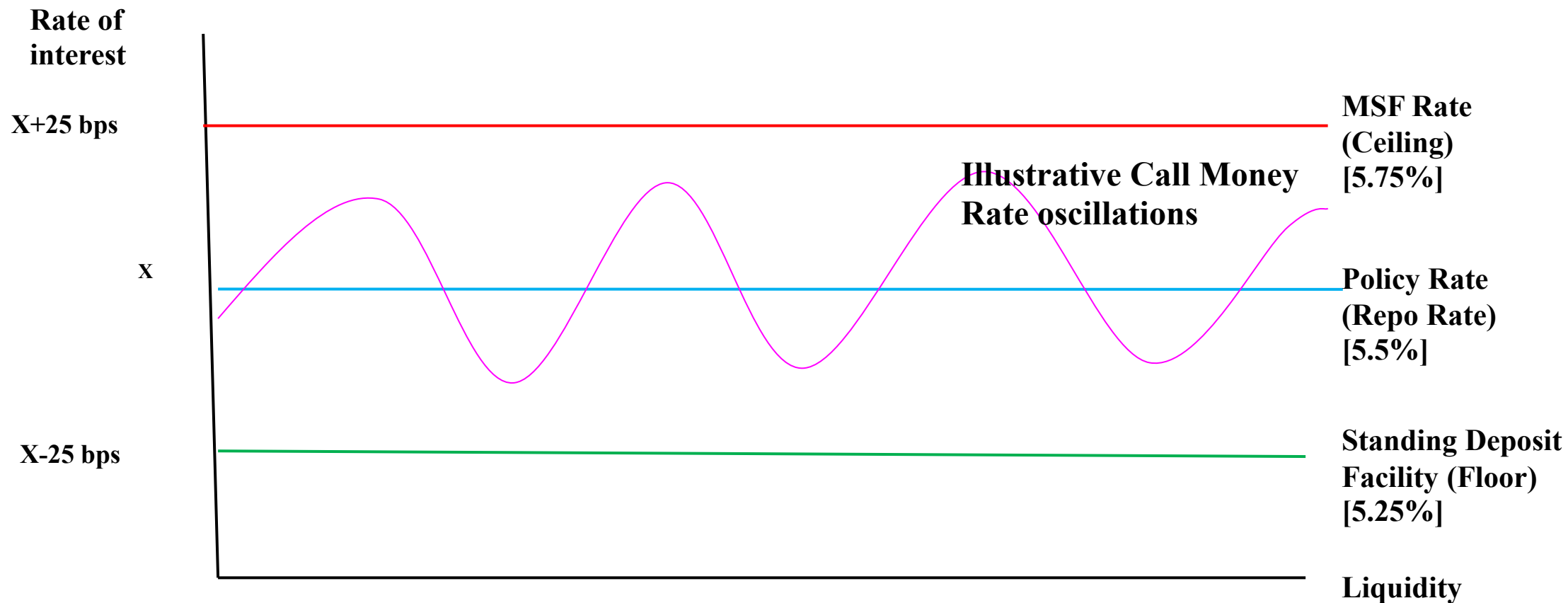
Money Market Mutual Funds (MMMFs)

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- **Money Market Mutual Funds (MMMF):** They are open-ended mutual funds which invest in short-term, low-risk money market (debt) instruments such as T-bills, CPs, CDs with an aim to benefit from lower volatility of short-term instruments so as to maintain stable net asset value (NAV).
- MMMFs originated in the U.S. in 1972 as largely to work around the **Regulation Q** that limited the interest rates depository institutions were allowed to pay depositors. As interest rates rose in the 1970s, Regulation Q gave depositors an incentive to find short-term investments outside of the banking system, such as T-bills, CPs & CDs. MMMFs offered consumers the ability to invest in these instruments. The number of MMMFs in the U.S. grew rapidly to 36 in 1975, 90 in 1980, 649 in 1990. Thereafter, on the back of money market defaults (in CPs in 1989 & 1990 and in derivatives in 1994), consolidation (M&A) and strict regulatory reforms followed (SEC's **Rule 2a-7** mandated stricter liquidity and credit quality limitation; & post GFC floating NAV which requirement by SEC). Consequently, the number of MMFS has dropped to prompted many MMMFs to shift from prime funds to govt only funds) the number of MMFs in the US has dropped to about 250 currently. However, their total assets has continued to grow and total over US\$8 tn today.
- In India. MMMFs were started in 1992 & they were brought under SEBI regulation in 2000.
- Since MMMFs are not banks, they have grown outside the sphere of central banks regulation and supervision and generally carry greater default risks than a bank or even an NBFC. In case interest rates rise abruptly they can book losses on their portfolio of money market securities (CPs, CDs whose price falls like that of fixed income securities like bonds). However, their investments in overnight or very short-term instruments offer offset.

Monetary Policy sets the yield for most money market instruments; as they hug the policy repo rate within an interest rate corridor

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Q to discuss: Why do the money market rates get anchored to near policy rate within the interest rate corridor, with odd breaches of the corridor on some occasions?

Money Markets often give the first sign of trouble brewing in the markets; act as leading indicator of institutional or systemic failures

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- Money markets act as the plumbing of the global financial system. These markets show signs of strain, often as a leading indicator of systemic financial stress - signalling through liquidity hoarding, credit tightening, or volume freezes indicating lack of trust between financial institutions:
- **Elevated Short-Term Rates:** Unusual spikes in overnight rates (e.g., SOFR, SONIA, ESTR) above a central bank's target rate (like the WACR or Interest on Reserve Balances-IORB) can point to an unexpected shortage of cash and heightened competition for funding
- **Widening Interest Rate Spreads:** The difference between risk-free reference rate that are often risky unsecured short-term lending rates (such as the earlier benchmarks LIBOR/Euribor or their new replacements such as SOFR, SONIA, ESTR) and the sovereign risk-free rates [such as G-sec benchmarks, overnight indexed swaps (OIS) or Treasury bills (T-bills) rates] measures the perceived risk of bank defaults. Widening spreads show that banks are charging a premium to lend to one another, signalling credit fears.
- **Shift to Safe Assets:** During stressed periods, demand surges for ultra-safe instruments like T-bills. Investors may also pull funds out of private money market funds if they fear defaults.
- **Central Bank Facility Usage:** A sudden spike in reliance on standing repo facilities indicates or discount windows indicate that financial institutions are struggling to secure necessary funding from private counterparties.

What are Debt Markets?

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BONDS VS. STOCKS

BONDS:



- Issues of debt
- Debt that is made with an investor for cash in exchange for payouts of interest
- Typically traded over the counter (OTC)
- Generally lower risk, lower reward
- Since 1929 have earned around 6% each year
- Can be made as corporate, municipal, or treasury bonds

STOCKS:



- Issues of a stake of ownership in a company
- A claim to a company's assets and earnings that often gives the investor voting rights and pays dividends
- Typically traded through a central exchange (like NYSE)
- Generally higher risk, higher reward
- Since 1929 have earned around 10% each year
- Are issued by companies at a stock exchange as IPOs

- The **Capital Market** = Equities Market + long-term debt >1 year tenor).
- Equity Markets trades in equity securities, or stocks. The distinctive characteristics of an equity is that it provides an buyer of the stock an ownership stake in a company which issues the stock. It entitles them to dividends and capital appreciation of the stock if the company performs well with risk that he may lose its value if the company's performance deteriorates.
- In contrast, bondholders lend money to company and gets coupon or interest income.
- **The Debt Market** is a market for the issuance, trading and settlement in debt instruments of various types. **Debt instruments are also called fixed income securities.** These are issued (i) Central government, (ii) State Governments, (iii) Municipal corporations, (iii) statutory corporations of Centre or states, (iv) banks, (v) other financial institutions, including NBFCs and (vi) non-financial companies.

Bond Basics... (1): What is a Bond?

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- Bonds are **fixed income** instruments that are issued and traded in debt capital markets as securities with maturities longer than one year. This differentiates them from money market securities.
- Bonds have more intricate cash flow patterns than money market securities, which usually have just one cash flow at maturity. This makes bonds more involved to price than money market instruments, and their prices more responsive to changes in the general level of interest rates.
- There are a large variety of bonds. The most common type is the *plain vanilla* (or *straight*, *conventional* or *bullet*) bond. This is a bond paying a regular (annual or semi-annual) fixed interest payment or *coupon* over a fixed period to maturity or *redemption*, with the return of principal (the par or nominal value of the bond) on the maturity date. All other bonds are variations on this.
- A Bond is a contractual agreement between the issuer and the bondholders. There are three elements that an investor needs to know about when investing in a bond:
 - The bond's features, including issuer, maturity, par value, and coupon rate and frequency, and currency denomination. These features determine the bond's cash flows and therefore are key determinants of investor's expected and actual return
 - The legal, regulatory and tax considerations that apply to the contractual agreement between the issuer and bondholders
 - The contingency provisions that may affect the bond's scheduled cash flows. These contingency provisions are options; they give the issuer or bondholders certain rights affecting the bond's disposal or redemption.

Bond Basics... (2): Some basic concepts related to Bonds?

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- **Maturity date** of a bond refers to the date when the issuer is obligated to redeem the bond by paying the outstanding principal amount. The **tenor** is the time remaining until bond's maturity date
- What is a **Bond's Par Value**? The **principal** of a bond is the amount the issuer agrees to repay the bondholders on the maturity date. This amount is also called the *par value, face value, nominal value, redemption value or maturity value*
- What is a **Bond's Coupon**? The **coupon rate** of a bond is the interest rate that the issuer agrees to pay each year until the maturity date. The annual amount of interest payments made is called the **coupon**. Coupon payments can be made annually, semi-annually, quarterly or monthly.
- A **plain vanilla bond** pays a fixed rate of interest. The coupon payment does not change during the life of the bond. A **Floating Rate Notes (FRNs) or Floaters** pay a floating rate of interest. The coupon rate of a FRN includes two components – a reference rate plus a *spread*. The spread is typically constant and expressed in *bips*(basis points). A *bip* (basis point) is equal to 0.01%. For example, an Indian FRN may have the rate specified as *MIBOR/ SORR + 0.50%*, which means that the coupon rate for this bond would be prevailing *Mumbai Inter Bank Offer Rate/ Secured Overnight Financing Rate plus 0.50%*. MIBOR is a collective name for a set of rates covering different lending offers among banks for different maturities ranging from overnight to one year. As part of the international benchmark reforms, RBI since Dec 2024 MIBOR with SORR and this alignment is underway. A **Zero-Coupon Bond** do not pay interest, instead they are issued at a discount to par value and redeemed at par value.

Bond Basics... (3): Yield to Maturity (YTM)

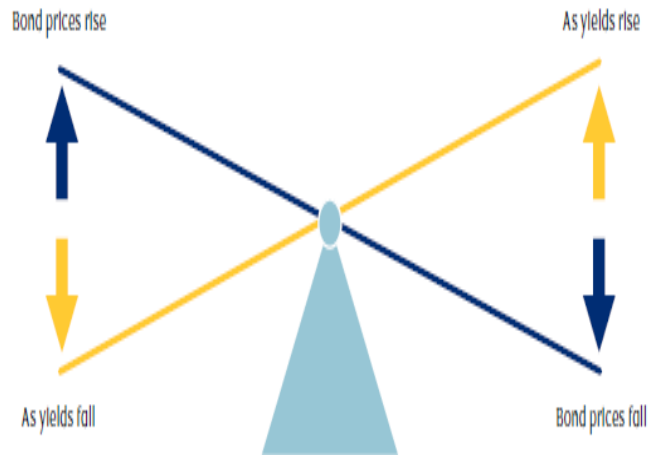
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- The **'current yield' of a bond** is equal to bond's annual coupon divided by the bond's price, expressed as a percentage.
- The **'Yield to Maturity (YTM)'** or 'Yield to Redemption' or 'Yield to call' is the internal rate of return on a bond's expected cash flows – that is the discount rate that equates the present value of bond's expected cash flows until maturity with the bond's price.
- There is an **inverse relationship between a bond's price and its YTM**, all else being equal. That is, the higher the bonds YTM, the lower its price. Thus, investors anticipating a lower interest rate environment (in which investors demand lower YTM) hope to earn a positive return from price appreciation
- **Duration** of a bond is a way to measure how sensitive bonds are to changes in interest rates. It is expressed as a number of years because it's a calculation of how long it will take the investor to be paid back on their loan in full. The further away a bond is from its maturity date, the longer its duration usually is – meaning it is more exposed to changing interest rates.

Bond Basics... (4): What are the risks an investor faces when investing in bonds?

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Bond prices and interest rates generally move in opposite directions



- There are two main risk bond investors should keep in mind:
- **Credit risk:** The risk that the issuer may default and not pay the principal and interest.
- Bonds issued by governments often pay a lower rate of interest than corporate bonds. This is because governments in general have the highest likelihood of repaying investors.
- **Interest rate risk:** The vulnerability of a bond to changes in interest rates. Usually, the longer a bond is from maturity, the greater interest rate risk it has because there is more uncertainty over the path of rates over time.

Bond Basics... (5): Duration of a Bond

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$$D = \frac{\sum (\text{present value of cash flow} \times \text{time to cash flow})}{\sum (\text{present value of cash flow})}$$

Mathematically this is written as (2.26) below.⁵

$$D = \frac{\sum_{n=1}^N \frac{nC_n}{(1+r)^n}}{P}$$

where

D is the Macaulay duration

C is the bond cash flow at time n

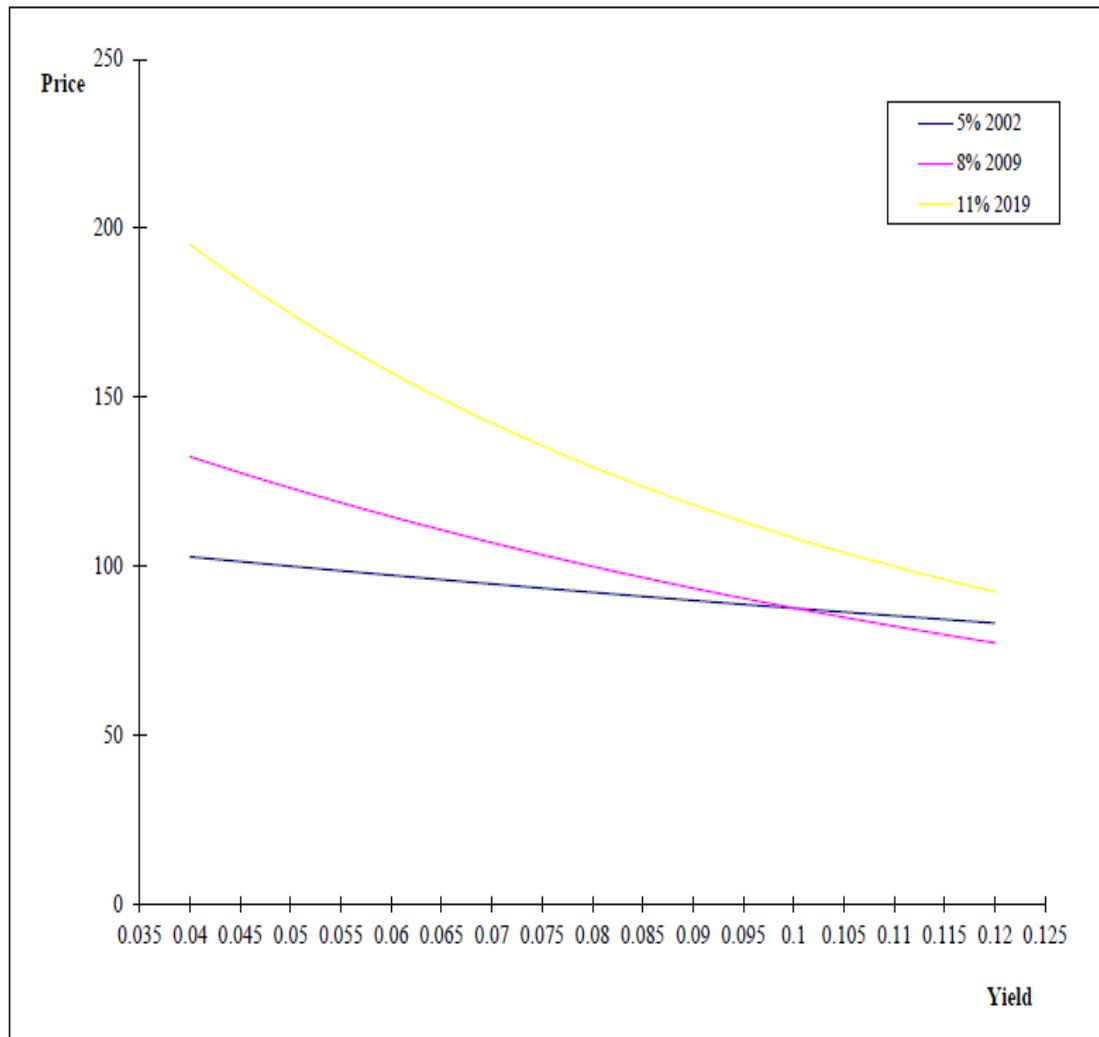
P is the present value of the n -period cash flow, discounted at the current yield to maturity

n is the number of interest periods.

- **Duration of a bond** is a way to measure how sensitive bonds are to changes in interest rates. It is expressed as a number of years because it's a calculation of how long it will take the investor to be paid back on their loan in full. The further away a bond is from its maturity date, the longer its duration usually is – meaning it is more exposed to changing interest rates. So, by changing average maturity of one's bond portfolio, one may be able to effect changes in duration.
- A higher duration means a bond price is more volatile in response to interest rate changes.
- Duration is affected by a bond's maturity, coupon payments, and yield, with longer maturities and lower coupon payments generally resulting in higher durations.
- Duration is important for portfolio management as the treasuries and CFOs seek to immunize themselves against interest rate risk or to optimize returns amid risk-return trade offs.

Bond Basics... (6): Convexity of a Bond

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- **Bond Convexity** measures the non-linear relationship between bond's price and its yield. It indicates how much a bond's duration changes as interest rates change, with high convexity indicating price gains from falling interest rates are greater than price losses from rising interest rates of same magnitude.
- Most bond portfolios are dominated by bonds with positive convexity, implying investors stand to benefit with falling interest rate environments.
- Duration assumes a linear relationship between a bond's price and interest rate changes, but this is an approximation. Convexity accounts for actual curved, or convex relationships.
- Positive convexity means that as interest rates fall, bond duration increases. This accelerates the bond price increase.

Bond Basics... (7): Coupon Payment Structures of Bonds

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- **Fixed Rate** structure promises a fixed rate of coupon paid on due date
- **Floating Rate** structure will not have a fixed coupon, instead their coupon rate is linked to an external reference rate, such as MIBOR
- **Step-up Rate** structures may be fixed or floating, but increases by specified margins at specified dates.
- **Credit-linked Rate** structure has a coupon that changes when the bond's credit rating changes
- **Payment-in-Kind** structure allows the issuer to pay interest in the form of additional amounts of the bond issue rather than as a cash payment
- **Deferred Coupon Rate** structure or Split Coupon Rate structure pays no coupons for its first few years but then pays a higher coupon than it otherwise normally would for the remainder of its life.
- **Bullet Bond** is a plain vanilla bond with one coupon payment.
- **Amortizing Bond** makes periodic payments of interest and repayments of principal. It is a **fully amortized** bond if these fixed periodic payments reduces the bond's outstanding principal amount to zero by the maturity date. If not, it is a **partially amortized bond** that will have a **balloon (large) payment** at maturity to retire bond's outstanding principal amount.
- **Perpetual Bond** will have coupon payments perpetually with no retirement date.

Bond Basics... (8): Some Types of Bonds

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- **Fixed Rate** structure promises a fixed rate of coupon paid on due date
- **Secured Bonds** are backed by collaterals to ensure debt repayment in case of default.
- **Collateral Trust Bonds** are secured by securities such as common shares, other bonds and other financial assets.
- **Equipment Trust Certificates** are bonds secured by specific types of equipment or physical assets, such as aircrafts, oil rigs etc.
- **Covered Bond** is a debt obligation backed by a segregated pool of assets called a cover pool. It is important to understand the risk in covered bonds even though they appear to be bankruptcy-protected. During the GFC there were no crashes of covered bonds
- **Bearer Bond** is a bond for which ownership is not recorded; only the clearing system will know who the bond owner is at a given time. In contrast, ownership is recorded in case of a **registered bond**

Who regulates the Capital Markets in India? ... (1)

SEBI as regulator of equity and corporate debt market in India

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SEBI is securities market regulator in India and has powers to regulate equities & corporate bonds

- The Securities and Exchange Board of India (SEBI) was set up as non-statutory body in 1988 and was later established as a statutory body in the year 1992 with the Securities and Exchange Board of India Act, 1992 coming into force on 30th Jan 1992.
- The Preamble of the SEBI describes its function as "...to protect the interests of investors in securities and to promote the development of, and to regulate the securities market and for matters connected therewith or incidental thereto"
- Listed securities in India are governed by three main laws: the Securities and Exchange Board of India Act 1992 (the SEBI Act); the Securities Contract (Regulation) Act 1956 (SCRA); and the Depositories Act 1996 (the Depositories Act), along with the rules, regulations and other subordinate legislations issued under these laws.
- The SCRA primarily regulates contracts and transactions in securities including derivatives and specifically provides for registration and regulation of market infrastructure institutions like recognised stock exchanges and clearing corporations in India. The Depositories Act deals with registration and regulation of securities depositories and specifically provides for the rights and obligations of various stakeholders, including the issuers of securities and depository participants who deal in such securities, as well as their legal and beneficial owners.
- The term 'securities' is defined under the SCRA and includes shares, scrips, stocks, bonds, debentures, debentures stock (and such other marketable securities), derivatives, government securities (such as treasury bills), units of collective investment schemes, mutual funds and pooled investment vehicles, as well as rights or interest in such securities.

Who regulates the Capital Markets in India? ... (1)

RBI as regulator of government bond market in India

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RBI is the central bank of the country and also a regulator of banks, NBFCs, money, FX & G-sec markets

- The Reserve Bank of India (RBI) was established on April 1, 1935 as the central bank of India. It was founded based on the recommendations of the Hilton Young Commission with its statutory basis as the RBI Act of 1934. The RBI initially commenced operations in Calcutta (now Kolkata) before permanently relocating to Mumbai in 1937. Initially a private entity, the RBI was nationalized in 1949.
- The Preamble of the RBI describes its function as “... it is expedient to constitute a Reserve Bank for India to regulate the issue of Bank notes and the keeping of reserves with a view to securing monetary stability in [India] and generally to operate the currency and credit system of the country to its advantage;
 - AND WHEREAS it is essential to have a modern monetary policy framework to meet the challenge of an increasingly complex economy;
 - AND WHEREAS the primary objective of the monetary policy is to maintain price stability while keeping in mind the objective of growth;
 - AND WHEREAS the monetary policy framework in India shall be operated by the Reserve Bank of India;].
- Apart from the RBI Act, RBI derives its functions from:
 - Banking Regulation Act, 1949 (powers to regulate and supervise commercial banks; & cooperatives (along with other Acts) while powers to regulate and supervise NBFCs are drawn from Chapter III B of the Reserve Bank of India Act, 1934
 - RBI derives its powers to regulate and supervise the G-Sec market from several acts, including the Reserve Bank of India Act, 1934 and the BR Act, 1949, additionally, through amendments to the SCRA, 1956 and most importantly through the enactment of the Government Securities Act, 2006 (GS Act)
 - powers to regulate the money market comes from Section 45W of the RBI Act, 1934, while powers to regulate FX market largely comes from the Foreign Exchange Management Act, 1999 (FEMA). The Payment and settlement Systems Act, 2007 gives it powers to regulate and supervise payment and settlement systems in India.

Ratings of the Bonds: What are its Key Components

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- Credit risk is the risk of loss that a lender may face due to the borrower's failure to repay debt.
- There are two key components of credit risk:
 - Whether the debt is likely to be paid on time? → Usually referred to as Probability of Default (PD)
 - In case of a default, how much loss will the lender incur? → Usually referred to as **Loss Given Default (LGD)**
- Conventional credit ratings assess the PD of the issuer.
- For credit risk assessment, it is important to have clarity on the definition of default.
 - CRAs: Typically follow a default definition of delay of even one day of even one rupee.
 - Banks and FIs: Typically follow a definition of delay beyond 90 days (tied to NPA recognition).
- In the rest of the presentation, we will cover the conventional credit ratings that address PD.

An important utility of Credit Rating

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- Ratings are used to determine risk weights for bank exposures
 - Regulatory capital requirement as per the risk weight is also factored in the pricing of loans

Rating category	AAA	AA	A	BBB	BB and
Risk weight	20%	30%	50%	100%	150%

The above table shows the linkage between external ratings and risk weight prescribed by the Reserve Bank of India (RBI), the Indian banking regulator.

Rating Scales in India

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Long-term rating scale – 20 points

Rating scale	Rating definition
AAA	Highest Safety
AA+	High Safety
AA	
AA-	
A+	Adequate Safety
A	
A-	
BBB+	Moderate Safety
BBB	
BBB-	
BB+	Moderate Risk
BB	
BB-	
B+	High Risk
B	
B-	
C+	Substantial Risk
C	
C-	
D	Default

Short-term rating scale – 9 points

Rating scale	Rating definition
A1+	Very Strong Safety
A1	
A2+	
A2	Strong Safety
A3+	
A3	
A4+	Adequate Safety
A4	
D	
	Minimal Safety
	Default

Outlook

Scale	Definition
Positive	Upward rating movement expected in 12-18 months
Stable	Rating expected to remain stable for next 12-18 months
Negative	Downward rating movement expected in 12-18 months

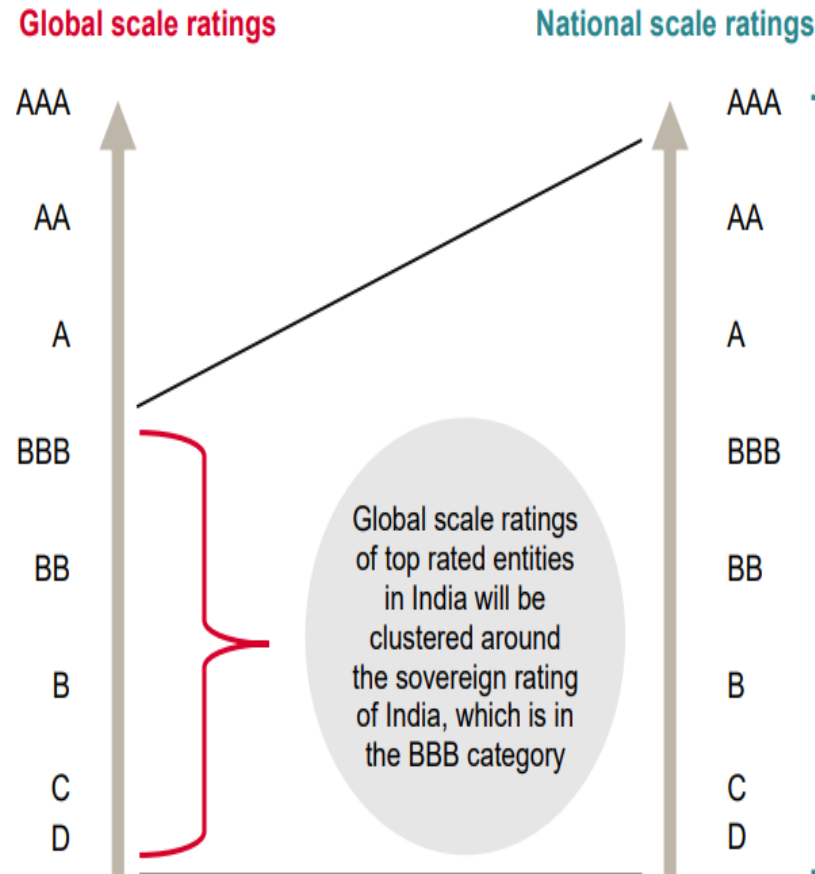
- The above scales are the standardised scales prescribed by the Securities and Exchange Board of India (SEBI), the regulator for Indian capital markets and Indian CRAs.
- Outlook' indicates the direction of the long-term rating over the next 6-12 months. Indian CRAs started assigning outlooks almost 17 years after they began operations. Outlooks were made mandatory by the regulator almost 30 years after CRAs started operating in India.

Global vs National Rating Scales ...(1)

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Global scale

- **Global CRAs** use this scale to **rate entities around the world**
- Global ratings are all **comparable**
- Useful for investors and lenders who can invest/lend across countries
- **Country risk** is an important parameter for arriving at the global rating of an entity. The entity rating is usually close to or less than the sovereign rating
- Global rating agencies: **S&P Global Ratings, Moody's, Fitch**, etc.
- The global rating scale may not be able to differentiate much between entities in a lower rated sovereign



National Scale

- **National CRAs** use this scale to rate entities in **their own country**
- National ratings of two different countries are **not comparable**
- Useful for investors and lenders who are constrained to hold investments/lend to issuers within a particular country
- Country risk is not included for assigning ratings on the national scale as all entities in a country face similar economic conditions
- Indian local rating agencies: **Crisil Ratings, ICRA, CARE**, etc.
- The national rating scale provides better differentiation between entities within the country

Global vs National Rating Scales ...(2)

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Country	Entity	Global scale rating	Domestic scale rating
India - S&P sovereign rating 'BBB'	Reliance Industries Ltd	BBB+	Crisil AAA
	State Bank of India	BBB	Crisil AAA
	Infosys Ltd	A	Crisil AAA
	Tata Power	BBB	Crisil AA
Argentina - S&P sovereign rating 'CCC-'	Banco Patagonia	CCC-	raCCC+
Taiwan - S&P sovereign rating 'AA+'	Bank of Taiwan	AA	twAAA
	Taiwan Semiconductor Manufacturing Co Ltd	AA-	twAAA
	Chang Hwa Commercial Bank Ltd	A	twAA+
	Nan Shan Life Insurance Company Ltd	BBB+	twAA
Uruguay - S&P sovereign rating 'BBB+'	Metlife Seguros	BBB+	uyAAA
	Banque Heritage (Uruguay)	BB-	uyA-
Canada - S&P sovereign rating 'AAA'	Arch Insurance Canada Ltd	A+	A+
	Aviva Insurance Company of Canada	AA-	AA-

Ratings Impact the yield of a Corporate Bond

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Illustration of an average Yield Spread over G-sec yield of similar maturity (not current yield spread)

PSUs	6-months	1 year	2 year	3 year	5 year
AAA	0.51 %	0.48 %	0.37 %	0.31 %	0.26 %
AA+	0.81%	0.78%	0.65%	0.61%	0.56%
AA	1.11%	1.08%	0.96%	0.96%	0.86%
AA-	1.36%	1.33%	1.21%	1.21%	1.11%

Yield on the bond used
for valuation



Yield on the G-Sec of the
similar duration



Credit spread for the corresponding
rating and the tenor as per the matrix

Example: 3-year “AA+” rated bond

3- year G-sec yield	Spread for 3-year AA+ rated bond	Bond Yield
7%	0.61%	7% + 0.61% = 7.61%

Green Bonds

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- **Green bonds** refer to debt securities issued to raise capital earmarked for green projects. The exact definition, however, varies depending on what constitutes green projects. For example, the Green Bond Principles (GBPs), which were established in 2014 and are maintained by the International Capital Market Association (ICMA), provide guidelines and green project categories (ICMA 2021). The Climate Bonds Standards (CBSs), built on top of the GBPs by the Climate Bonds Initiative (CBI), provide a sector-specific definition of “green” and are used for the certification of green bonds by CBI (Climate Bonds Initiative 2019)
- The origin of the Green bonds can be traced to the bond issued by the European Investment Bank in 2007 as the first green bond. The Green Bond Principles introduced by the International Capital Market Association (ICMA) in 2014 provided a set of voluntary principles that promote transparent and standardized climate objectives and impacts that qualify for a green label. Increasingly originated by financial institutions, sovereigns, and nonfinancial corporates, green bond issuances have grown rapidly over the last decade, reaching more than US\$600 billion in 2021. Green bonds are one of the most important types of climate-related bonds, representing one-half of the global total issuances of climate-related bonds and about 80 percent of climate-related bond issuances in EMDEs
- Several sovereigns has issued green bonds since 2016.

Evolution of Green Bond Framework in India

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- Development of the Green Bond framework in India can be traced back to the the 'National Voluntary Guidelines on Social, Environmental and Economic Responsibilities of Business' (NVGs) which were issued by the Ministry of Corporate Affairs (MCA) in 2011 and the filing of the Business Responsibility Report (BRR) containing ESG (Environmental, Social and Governance) disclosures was first introduced by SEBI for the top 100 listed entities (by market capitalization) in 2012, through the then listing agreement as per the disclosure requirement emanating from the NVF+G of MCA.
- In December 2018, the National Guidelines on Responsible Business Conduct, 2018 (NGRBC), an improvement over the NVGs, were issued. In March 2015, The International Capital Market Association (ICMA) published a document called Green Bond Principles (GBP). The GBPs were a set of best practices for issuing green bonds.
- SEBI thought fit to fill the vacuum caused by lack of regulation which resulted in many Indian players tapping the overseas market for green / sustainable finance. SEBI initiated a consultation process for disclosure requirements for Public Issue and Listing of Green Bonds and listing of privately placed Green Bonds. A concept paper based upon the Green Bond Principles, 2015 of the ICMA was placed on the SEBI website on December 3, 2015, for seeking public comments. After taking public comments, SEBI issued its First Green Bonds Framework in May 2017. gaining from the experience it issued a New Green Bond Framework applicable for all green bond issuances since April 1, 2023

What is a Green Bond as per the Framework in India

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- The framework provides that a security would be considered 'green' if proceeds from its issue were to be utilised for project(s) and/or asset(s) falling under any of the following broad categories:
 - a) Renewable and sustainable energy including wind, solar bioenergy, other sources of energy which use clean technology, etc.
 - b) Clean transportation including mass/public transportation, etc.
 - c) Sustainable water management including clean and/or drinking water, water recycling, etc.
 - d) Climate change adaptation
 - e) Energy efficiency including efficient and green buildings, etc.
 - f) Sustainable waste management including recycling, waste to energy, efficient disposal of wastage, etc.
 - g) Sustainable land use including sustainable forestry and agriculture, afforestation etc.
 - h) Biodiversity conservation
- Plus, certain disclosures had to be made in the offer document as well as continually to the stock exchange by the issuer.

Important Modifications in the Green Bond Framework in India

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- Enhanced scope of definition of green debt security by including new modes of sustainable finance in relation to pollution prevention and control, eco-efficient products, etc.
- Introduced the concepts of:
 - blue bonds (related to water management and marine sector),
 - yellow bonds (related to solar energy) and
 - transition bonds as a sub-category of green debt securities
- Broadened scope now includes: renewable and sustainable energy including wind, bioenergy, other sources of , energy which use clean technology, clean transportation including mass/public transportation, climate change adaptation including efforts to make infrastructure more resilient to, energy efficiency enhancements, including efficient and green buildings, sustainable waste management including recycling, waste to energy, efficient disposal of wastage, sustainable land use including sustainable forestry and agriculture, afforestation, biodiversity conservation,

Financing Pattern of the Firms

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- In your Finances courses you would have already studied the issues relating to capital structure - the debt vs equity choices a firm has to make
- Modigliani-Miller (MM) proposition-I says that the value of an unlevered firm is the same as the value of the levered firm. So, the capital structure does not matter. It was based on the assumption that companies are operating in the world of perfectly efficient markets, they do not pay any taxes, the trading of securities is executed without any transaction costs, bankruptcy takes place without any bankruptcy costs, and perfect information symmetry prevails.
- Modigliani-Miller (MM) proposition-II states that the company's cost of equity is directly proportional to the company's leverage level. An increase in leverage level induces a higher default probability for a company. Therefore, investors tend to demand a higher cost of equity (return) to be compensated for the additional risk. However, tax shields may make the cost of equity less sensitive to the leverage level.
- Information asymmetry theory states that firms prefer internal finance (own funds, viz, equity) to external finance (borrowed funds viz bonds/ debentures, and long-term debt). The other way to look at this is that firms will choose retained profits (internal equity), raising fresh equity (external equity) and debt (borrowings) in decreasing order of preference for raising finance as controlling shareholders and management want to avoid scrutiny that comes with raising fresh equity and even more so in the case of bank borrowings.

Financing of the Corporates: The Pecking Order theory of Capital Structure

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- How do Corporates finance themselves? What are firms' sources (financing pattern) and uses of finance (investment & other spending)?
- In his Presidential Address to the American Finance Association Stewart Myres pointed out that the capital structure puzzle is far more complex than even the dividend puzzle. John Lintner's model of 1956 explains how firms set dividends despite the MM irrelevance but we do not know how firms choose the debt, equity or hybrid securities they issue & advanced the pecking order theory propounded in a seminal paper by Myers & Majluf (1984)
- Myers and Majluf (1984) pointed to the information asymmetry that exists in financial markets stating that management knows more about the true value of the firm than the investors and, therefore, they rely more on internal sources of finance (retained earnings) and prefer debt to equity if external finance is to be used.
- Fazzari, et al (1984) extended the argument of information asymmetry in capital markets and explained that a firm's internal cash flow may affect investment spending because of a "financing hierarchy" in which internal funds have a cost advantage over new debt or equity finance. Financial constraints in capital markets can magnify the macroeconomic effect of shocks to cash flow or liquidity that reduce some firms' access to low-cost finance and worsen their balance sheet positions.
- Ajit Singh (1995), showed that financing pattern of the firms in developing economies differ from those in developed countries as they rely more on external finance and more on external equity than debt finance.

Financing Pattern of the US Firms: Fazzari, Hubbard & Petersen, Brookings Papers of Economic Activity (1988)

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Steven M. Fazzari, R. Glenn Hubbard, and Bruce C. Petersen 147

Table 1. Sources of Funds, by Asset Class, U.S. Manufacturing Firms, 1970–84

<i>Firm size</i>	<i>Source of funds (percent of total)^a</i>				<i>Percentage of long-term debt from banks</i>	<i>Average retention ratio</i>
	<i>Short-term bank debt</i>	<i>Long-term bank debt</i>	<i>Other long-term debt</i>	<i>Retained earnings</i>		
All firms	0.6	8.4	19.9	71.1	29.6	0.60
<i>Asset class</i>						
Under \$10 million	5.1	12.8	6.2	75.9	67.3	0.79
\$10–50 million	5.9	17.4	6.9	69.8	71.6	0.76
\$50–100 million	3.1	12.9	5.3	78.7	71.0	0.68
\$100–250 million	–0.2	13.3	12.0	74.9	52.4	0.63
\$250 million–\$1 billion	–2.3	10.6	15.4	76.3	40.8	0.56
Over \$1 billion	–0.6	4.8	27.9	67.9	14.7	0.52

Source: Authors' calculations based on data taken from U.S. Department of Commerce, Bureau of the Census, *Quarterly Financial Reports of Manufacturing, Mining, and Trade Corporations*, various issues. The data underlying the calculations are expressed in 1982 dollars.

a. Funds raised from new equity issues are excluded from the calculations.

Financing Pattern of the Indian Firms

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Financing Pattern for a sample of 6,955 NGNF Public Ltd Cos (%)

	2022-23	2023-24
INTERNAL SOURCES	52.0	68.6
Paid-up capital	7.2	3.5
Other equity (mainly reserves & surplus)	19.5	38.8
Provisions (mainly depreciation)	24.8	26.2
Others (mainly deferred government grants)	0.4	0.1
EXTERNAL SOURCES	48.0	31.4
Share capital and premium	12.3	2.4
bank borrowings	7.3	5.3
Bonds & debentures	0.7	2.5
Commercial paper	(3.5)	(0.4)
Trade payables	4.2	4.5
Other current & non-current liabilities	27.1	17.1
TOTAL	100.0	100.0

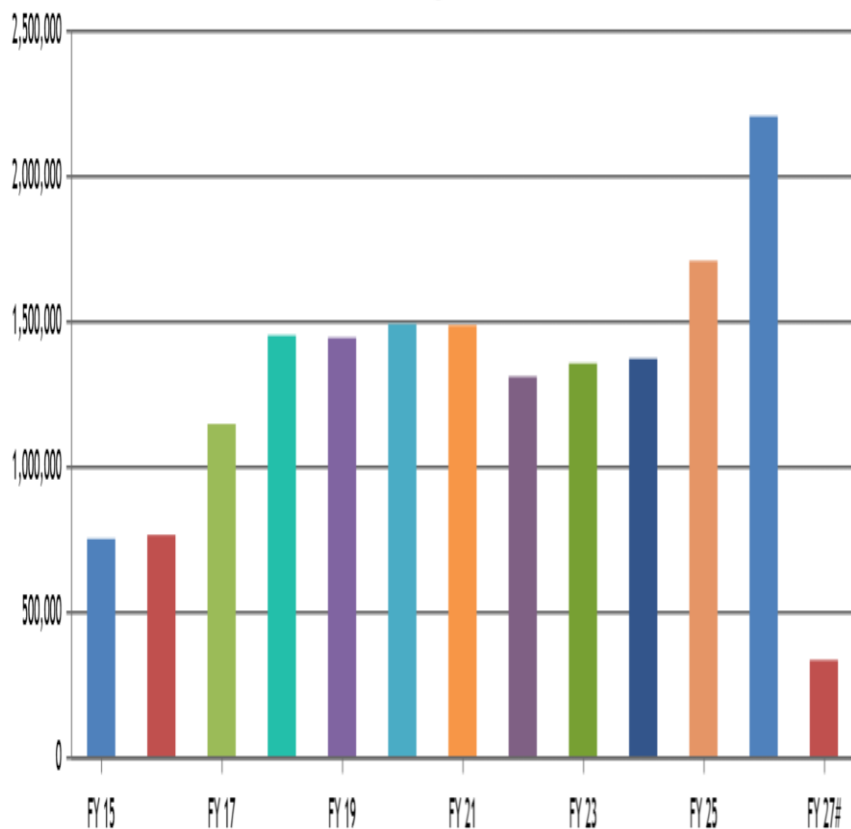
- Indian corporates have effectively deleveraged themselves. The leverage of NGNF public limited companies, as measured by debt-to-equity ratio, declined to multi-year low of 35.9 during 2023-24
- Interest coverage ratio (ICR) improved to 4.1 during 2023-24 as growth in gross profit outpaced the growth in interest expenses; ICR of manufacturing companies remained stable at 6.3, while it improved marginally to 3.2 for services companies
- The financing pattern of the Indian corporates has shifted markedly from the heavy reliance on external equity in the mid-1990s and now matches the pecking order theory with over half of the financing coming from internal sources – mainly reserves & depreciation provisions and in hierarchy external equity and bank borrowings follow with much smaller shares.
- Corporate bonds finance a very small share.

Trades in Corporate bond Markets in India

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In Rs cr; FY27 data is till May 2026

Trades In Corporate Bonds



- Globally, the corporate bonds outstandings were valued at ~\$36.4 trillion (lakh crores) at end-2025, with the U.S. and China accounting for the majority of outstanding debt. Inclusive of syndicated loans the market stands at In India, outstanding corporate bond volumes have grown significantly to ~₹59.2 (~\$625 billion) in May 2026, though this market remains smaller relative to the country's GDP compared to developed nations.
- Nevertheless, secondary trades in the Indian corporate bond markets have grown and touched a new high in FY26 with 28.42 lakh trades settled at a total amount of Rs 22.1 tn. A centralized database portal – Bond Central – has been recently launched by Online Bond Platform providers Association (OBPP) for reporting and enhanced transparency.
- The repo trades in corporate bonds settled through CCIL has, however, have fallen after FY24 and in FY26 some 10.8 thousand trades were settled with CCIL with an amount of Rs8.6 tn. The fall is also because of new entities coming into play. SEBI authorizing Limited Purpose Clearing Corporation (LPCC) and AMC Repo Clearing Limited (ARCL) expanding its tri-party repo services to allow mutual funds to choose their clearing corporation. However, this was a regulatory arbitrage as from the RBI Act, RBI draws powers to regulate repo in all securities. RBI has now authorised ARCL for repos in corporate bonds.

What is the Government Securities (G-sec) Market?

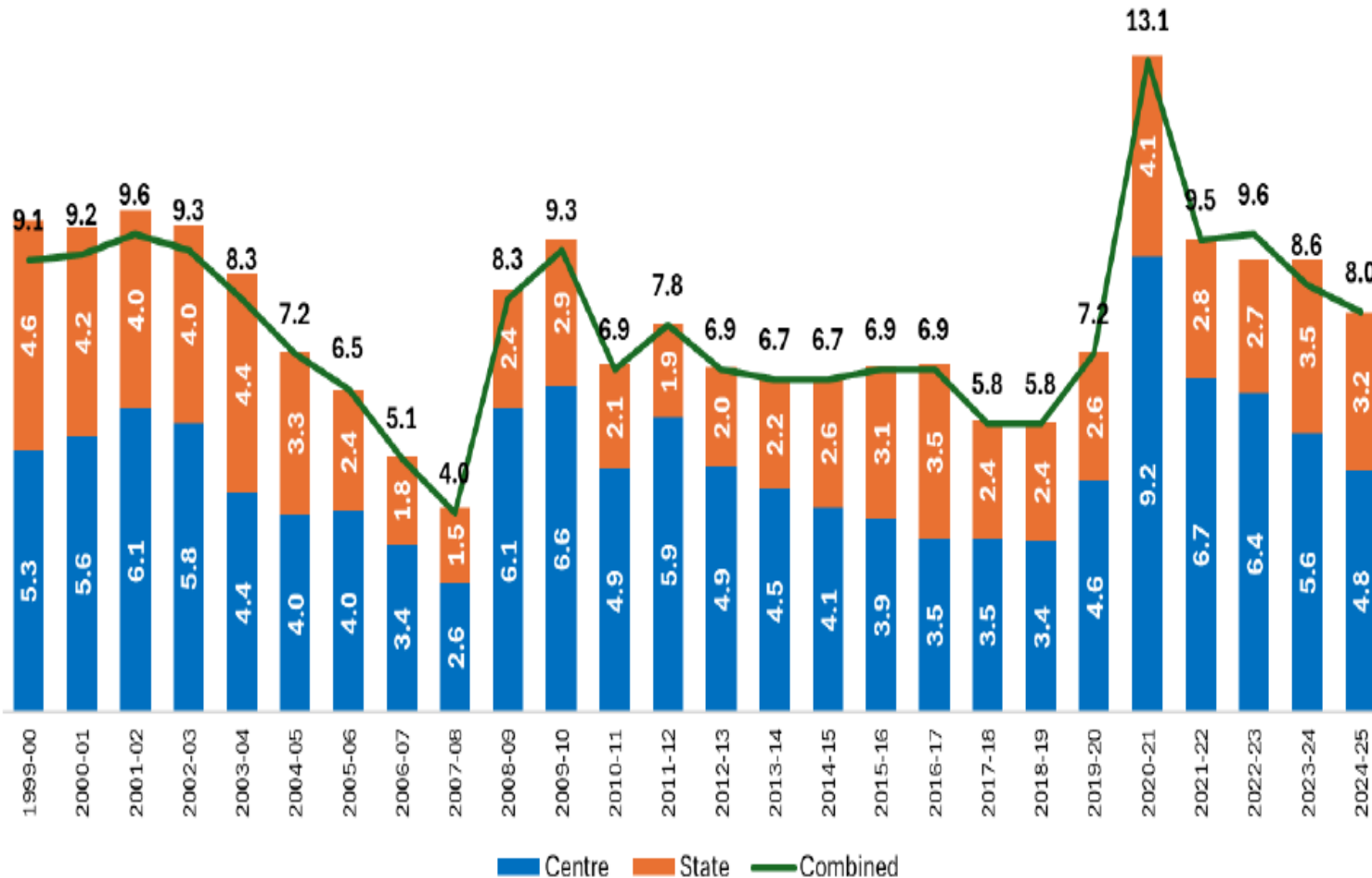
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- G-sec market is where debt instruments issued by the Government (both central and states) are traded for meeting its financing requirements. The first issuances of the G-sec are made in primary market, while subsequent buying and selling of the issued security is traded in the secondary market.
- The challenge before the G-sec market functioning is to ensure that the increased borrowing requirements of the Government, to finance high fiscal deficits is met in a cost-effective manner without distorting the financial system (by repressed yields rather than market clearing yields).
- The supply of government securities is generally exogenous to the market, determined mainly by the fiscal policy of the Government.
- From the perspective of an issuer, i.e., the Government, a deep and liquid government securities market facilitates its borrowings from the market at reasonable cost.
- As stipulated under the **Fiscal Responsibility and Budget Management Act, 2003**, the Reserve Bank has withdrawn from participating in the primary market for government securities from April 1, 2006.

G-sec Market exists because Governments run Fiscal Deficits

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Gross Fiscal Deficit of Centre and States (% of GDP)

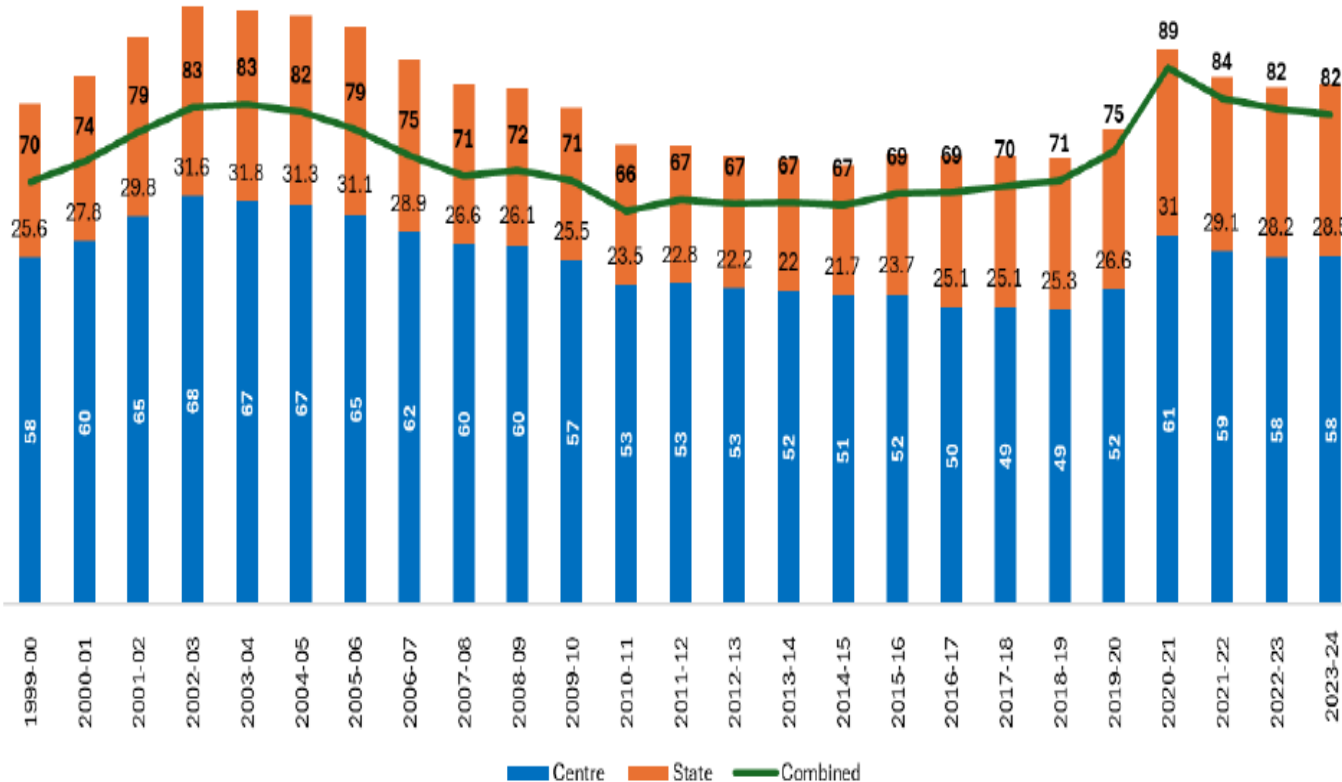


- Despite the sharp fiscal consolidation, combined deficit ratio remains above pre-pandemic levels
- GFD/GDP for Centre was 4.4% in FY26RE & 4.3% in FY26BE. State's fiscal deficit was 3.5% in FY25RE; 3.3% in FY26PA & 3.3% in FY27BE. So, its ratio remains more than double that of 1.5% in FY2008 & markedly more than 2.6% in FY20 (pre-pandemic).
- The adoption of the FRBM by the State governments beginning 2000s contributed to an improvement in their key fiscal parameter but several states breach the 3% ceiling

Because G-sec Market exists Governments pile up Public Debt

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Combined Public debt/liabilities of General Government(% of GDP)



- The Centre's public debt/ GDP ratio has declined from the pandemic high of 62.7% at end-FY21 to 49.1% at end-FY25 but remains well above 40% of reco of the FRBM Review Committee
- States public debt/ GDP ratio declined from 31.0% at end-FY21 to 28.5% at end-FY24; however, it remains well above the level of 20% reco of the FRBM Review Committee. Combined deficit was 13.1% in FY21 but has been brought down to 7.8% in FY25RE (FY26BE at the pre-pandemic level of 7.2%).
- India's central government liabilities-to-GDP ratio is estimated at 55.6% for the FY27BE, down from 56.1% in FY26RE and aligns with the government's medium-term fiscal consolidation target to reduce outstanding liabilities to around 50% of GDP by 2031.
- India's combined public debt/ GDP ratio has declined from 89.5% in FY21 to 81.% in FY25RE & 81.9% in FY26BE but remains above the threshold of debt sustainability should India's growth falls.

Market Borrowings of Centre and States

Rs. Crore							
Year	Gross Market Borrowings	Repayments	Net Market Borrowings	Gross Market Borrowings	Repayments	Net Market Borrowings	Gross Market Borrowings of centre and States
	Centre			All States			
2016-17	5,82,000	3,39,343	2,42,657	3,81,979	39,287	3,42,692	9,63,979
2017-18	5,88,000	2,64,957	3,23,043	4,19,100	78,819	3,40,281	10,07,100
2018-19	5,71,000	2,58,737	3,12,263	4,78,323	1,28,680	3,49,643	10,49,323
2019-20	7,10,000	4,36,150	2,73,850	6,34,521	1,47,067	4,87,454	13,44,521
2020-21	13,70,324	2,27,210	11,43,114	7,98,816	1,47,039	6,51,777	21,69,140
2021-22	11,27,382	2,64,279	8,63,103	7,01,626	2,09,143	4,92,483	18,29,008
2022-23	14,21,000	3,12,739	11,08,261	7,58,392	2,39,562	5,18,830	21,79,392
2023-24	15,43,000	4,40,129	11,02,871	10,07,058	2,89,918	7,17,140	25,50,058
2024-25	14,00,697	3,61,422	10,39,275	10,73,310	3,19,965	7,53,345	24,74,007
2025-26 (July 30, 2025)	5,53,000	1,35,086	4,17,914	3,24,368	1,00,550	2,23,817	8,77,368
	Growth (Per cent)						
2017-18	1.0	-21.9	33.1	9.7	100.6	-0.7	4
2018-19	-2.9	-2.3	-3.3	14.1	63.3	2.8	4
2019-20	24.3	68.6	-12.3	32.7	14.3	39.4	28
2020-21	93.0	-47.9	317.4	25.9	0.0	33.7	61
2021-22	-17.7	16.3	-24.5	-12.2	42.2	-24.4	-16
2022-23	26.0	18.3	28.4	8.1	14.5	5.3	19
2023-24	8.6	40.7	-0.5	32.8	21.0	38.2	17
2024-25	-9.2	-17.9	-5.8	6.6	10.4	5.0	-3

G-sec Market exists because Governments run Fiscal Deficits

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राजकोषीय घाटा वित्तपोषण के स्रोत Sources of Financing Fiscal Deficit

		(₹ करोड़) (In ₹ crore)			
		2024-2025 वास्तविक Actuals	2025-2026 बजट अनुमान Budget Estimates	2025-2026 संशोधित अनुमान Revised Estimates	2026-2027 बजट अनुमान Budget Estimates
1. ऋण प्राप्तियां (निवल)	1. Debt Receipts (Net)	1573823	1566452	1512770	1663066
i. बाजार उधार (सरकारी प्रतिभूति) *	i. Market Borrowings (G-sec) *	1074512	1153834	1040439	1173210
ii. अल्पकालीन उधार (राजकोषीय हुंडी आदि)	ii. Short term Borrowing (T-Bill etc.)	-160154	0	0	130000
iii. अल्प बचतों की तुलना में प्रतिभूतियां	iii. Securities against Small Savings	429502	343382	372192	386772
iv. राज्य भविष्य निधियां	iv. State Provident Funds	3999	5000	3800	3500
v. अन्य प्राप्तियां (आंतरिक ऋण निधि और लोक लेखा)	v. Other Receipts (Internal Debts and Public Account)	178692	40746	75877	(-)45801
vi. विदेशी ऋण	vi. External Debt	47271	23490	20462	15385
2. नकद शेष में आहरण द्वारा कमी	2. Draw Down of Cash Balance *	608	2484	45722	32702
कुल जोड़ (राजकोषीय घाटा)	Grand Total (Fiscal Deficit)	1574431	1568936	1558492	1695768

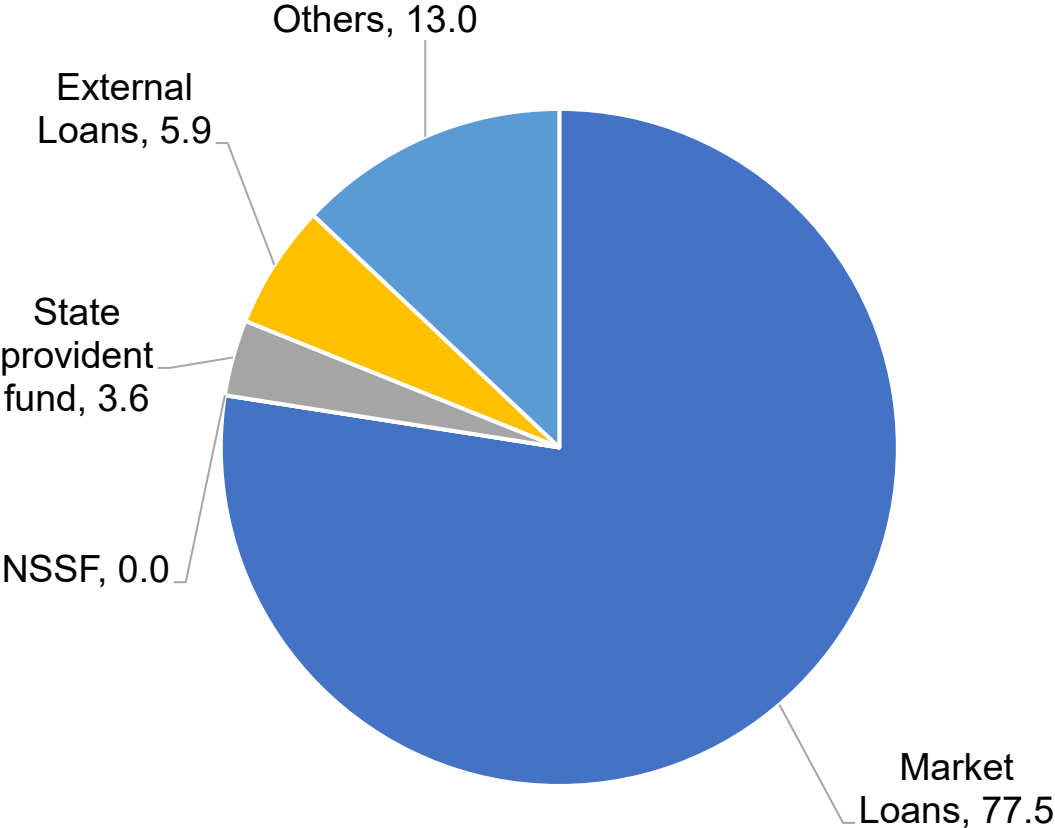
- * राशि पुनः खरीद और स्विचिंग प्रचालन को हटाकर है।
- * The amounts are net of Buyback and Switching operations.

- The Gross Market Borrowing for the Government of India for the financial year 2026-27 is estimated at ₹17.20 lakh crore. Out of this, the net market borrowing from dated securities is projected at ₹11.70 lakh crore, with the remaining balance funded through small savings and other sources
- The size of the government market borrowing which was rolling back with successive reductions in FY25BE & FY26BE, has widened in FY27
- Thus, government plans to borrow 69% of its fiscal deficit by issuing G-secs (dated securities)
- Small savings is budgeted to finance 22.8% of its GFD

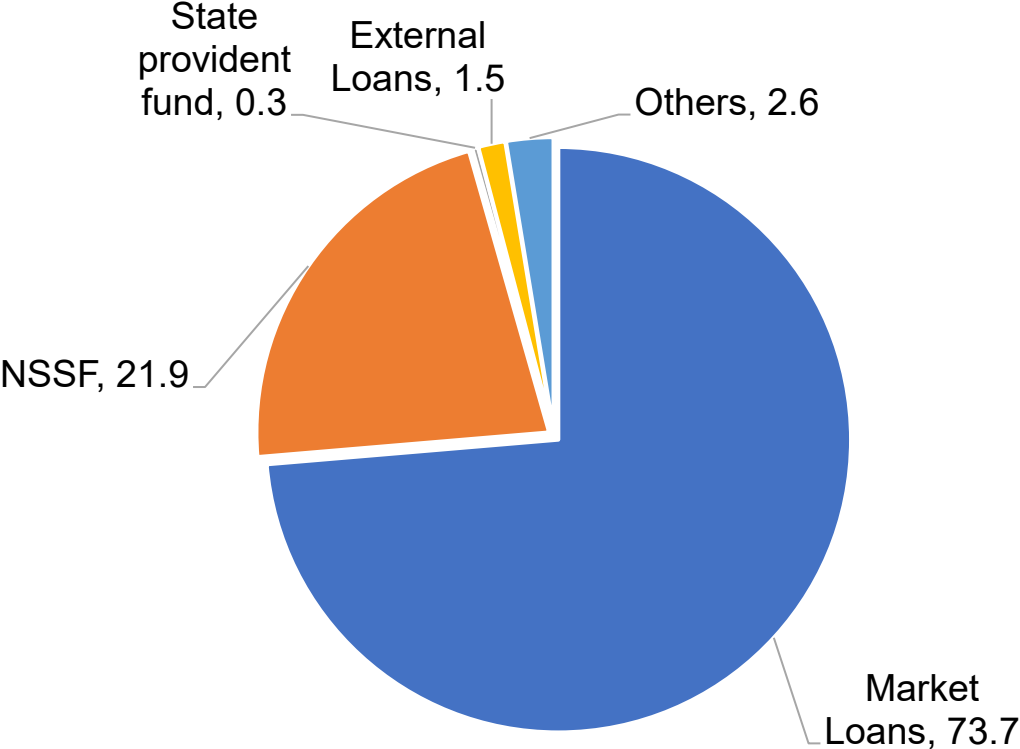
Market Borrowing dominates the financing pattern of GFD of the Central Government

43

2006-07

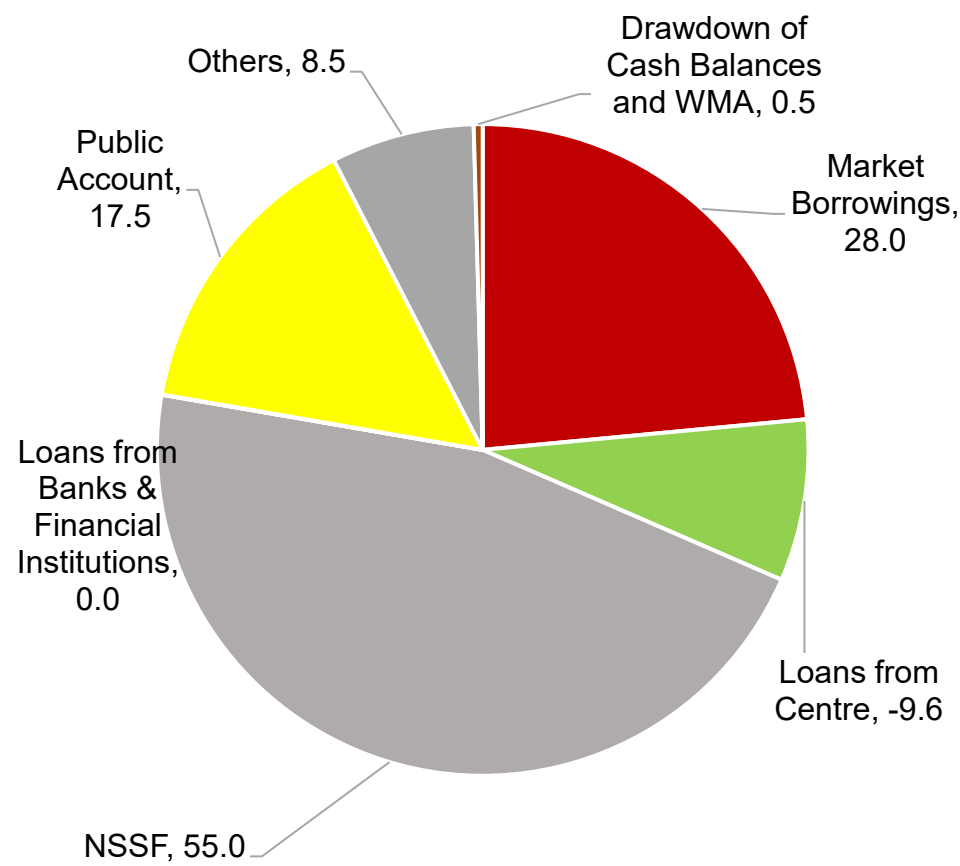


2024-25

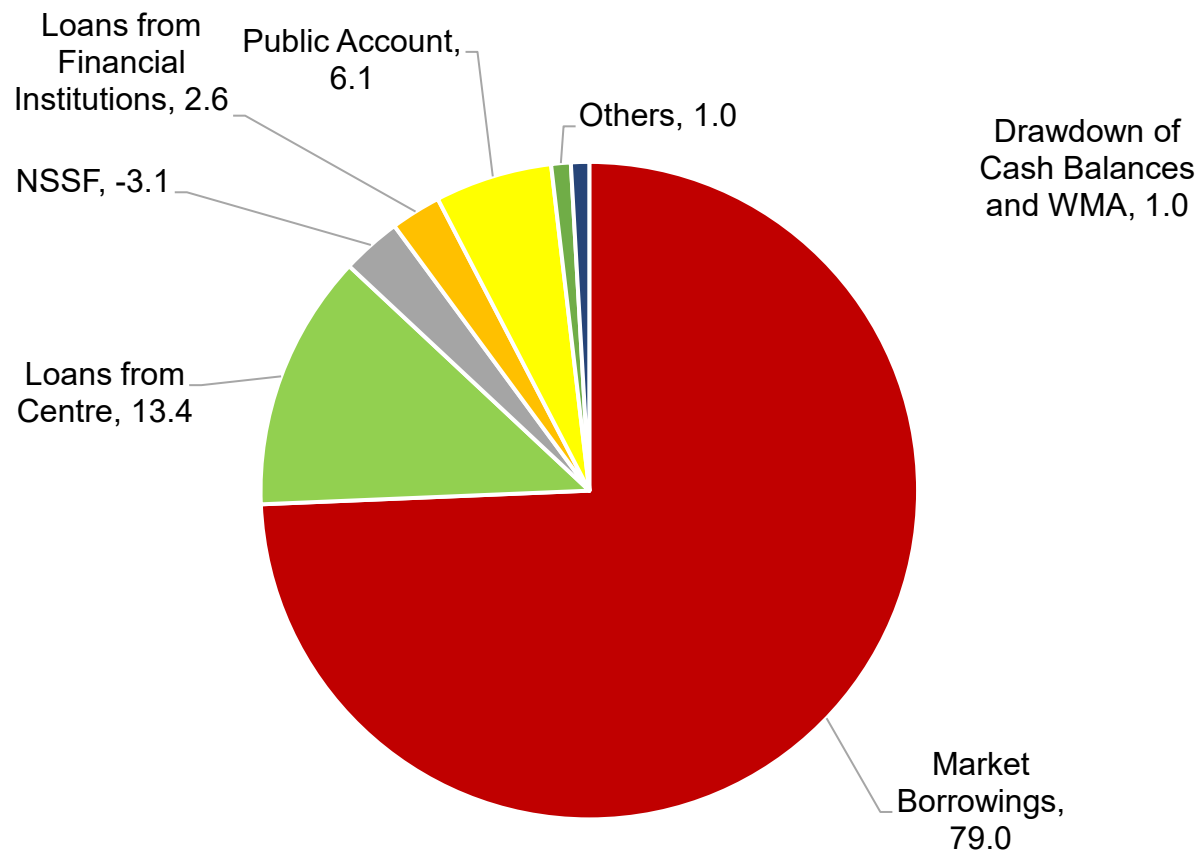


Market Borrowing has become important for financing GFD of the State Governments

2004-05



2024-25 BE



Institutional Structure of G-sec market in India

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- As the debt manager to the central and state governments, the Reserve Bank manages the government market borrowing programme in a non-disruptive manner while adhering to the broad objectives of cost optimisation, risk mitigation and market development.
- The Internal Debt Management Department (IDMD) of the Reserve Bank is entrusted with the responsibility of managing the domestic debt of the central government by statute vide Sections 20 and 21 of the Reserve Bank of India (RBI) Act, 1934, and of 28 state governments and two union territories (UTs) in accordance with respective bilateral agreements as provided in Section 21A of the RBI Act.
- Public Debt Management Cell, Budget Division, Department of Economic Affairs, Ministry of Finance, Govt. of India- Plan borrowings of the Gol, including market borrowings, other domestic borrowings, SGBs, manage Central Government's liabilities including NSSF, contingent liabilities.
- Each state government is responsible for their own debt servicing and redemption.
- The Public Debt Office (PDO) of the Reserve Bank of India acts as the registry / depository of G-Secs and deals with the issue, interest payment and repayment of principal at maturity.
- Clearing Corporation of India Ltd. (CCIL) - For secondary market operations (also through stock exchanges).

Operational Aspects of the Issuance of G-Sec ... (1)

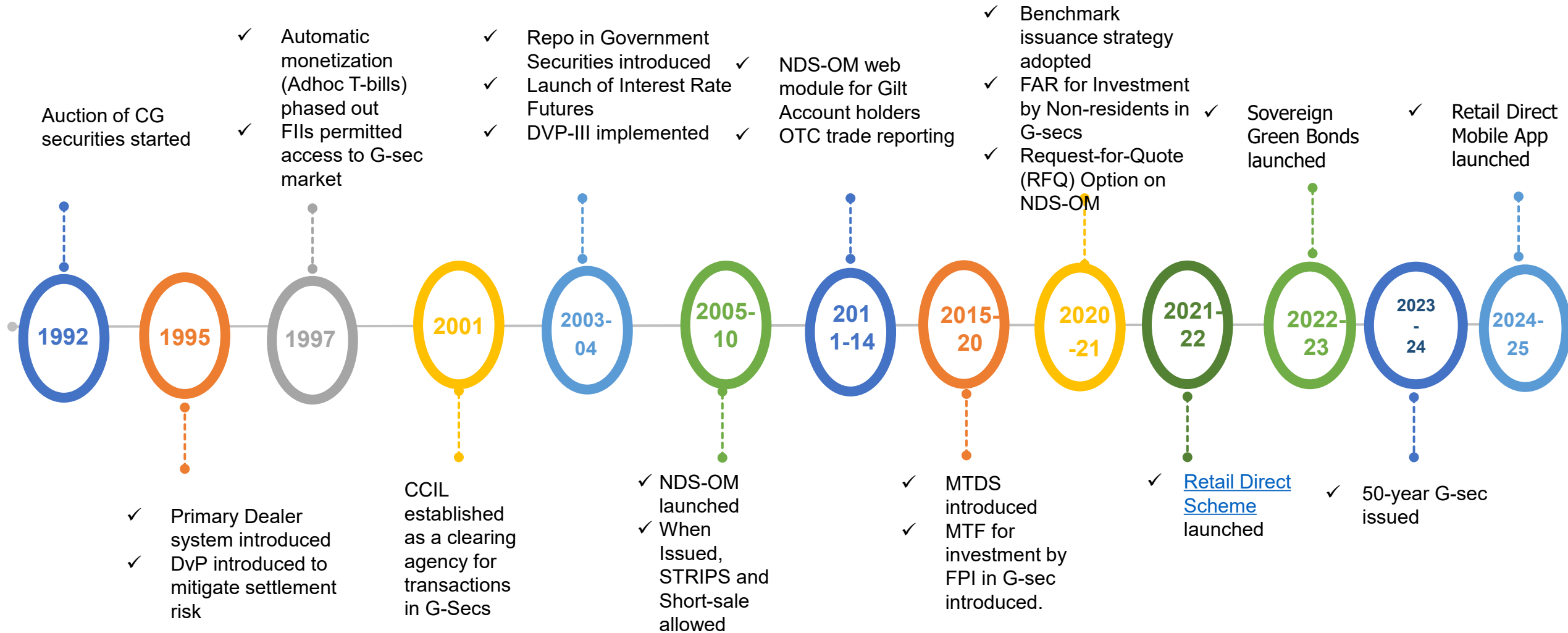
- Main Participants in the auction process- Commercial banks, scheduled UCBs, primary dealers (PDs), insurance companies and provident funds, who maintain funds account (current account) and securities accounts (Subsidiary General Ledger (SGL) account) with RBI.
- The results of the auction are published by RBI at stipulated time.
- All non-e-Kuber members including non-scheduled UCBs can participate in the primary auction through scheduled commercial banks or PDs (called as Primary Members-PMs).
- For this purpose, the UCBs/individuals etc. need to open a securities account with a bank / PD – such an account is called a Gilt Account. A Gilt Account is a dematerialized account maintained with a scheduled commercial bank or PD. The proprietary transactions in G-Secs undertaken by PMs are settled through SGL account maintained by them with RBI at Public Debt Office (PDO).
- The transactions in G-Secs undertaken by Gilt Account Holders (GAHs) through their PMs are settled through Constituent Subsidiary General Ledger (CSGL) account maintained by PMs with RBI at PDO for its constituent (e.g., a non-scheduled UCB).

Operational Aspects of the Issuance of G-Sec ... (2)

47

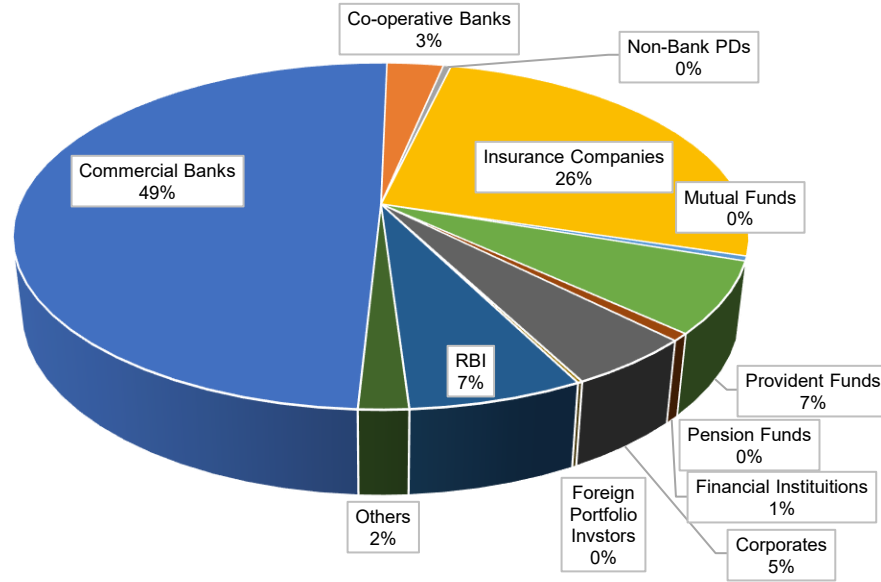
- Central Government (operationally, the RBI, in consultation with the GoI), issues an indicative half-yearly auction calendar which contains information about the amount of borrowing, the range of the tenor of securities and the period during which auctions will be held. A Notification and a Press Communique giving exact particulars of the securities, viz., name, amount, type of issue and procedure of auction are issued by the Government of India about a week prior to the actual date of auction.
- Central G-Sec auction- Every Friday, State Government G-Sec (SGSs) auction- Every Tuesday, T-Bill auctions- Every Wednesday to issue T-bills of 91day, 182 day and 364 day tenors.
- Settlement for the securities auctioned is made on T+1 day i.e. on the working day following the trade day.
- The Reserve Bank releases a quarterly calendar of T-bill and SGSs issuances for the upcoming quarter in the last week of the preceding quarter. e.g. calendar for April-June period is notified in the last week of March.
- The Reserve Bank of India announces the issue auction details of Central government, State governments securities and T-bills through a press release on its website every week.

Government Securities Market – Journey of Reforms

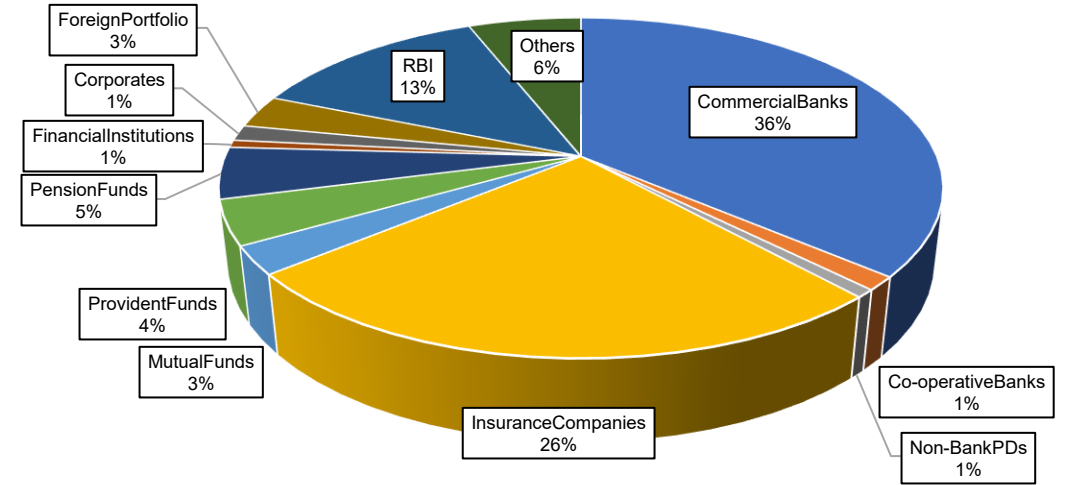


Ownership Pattern

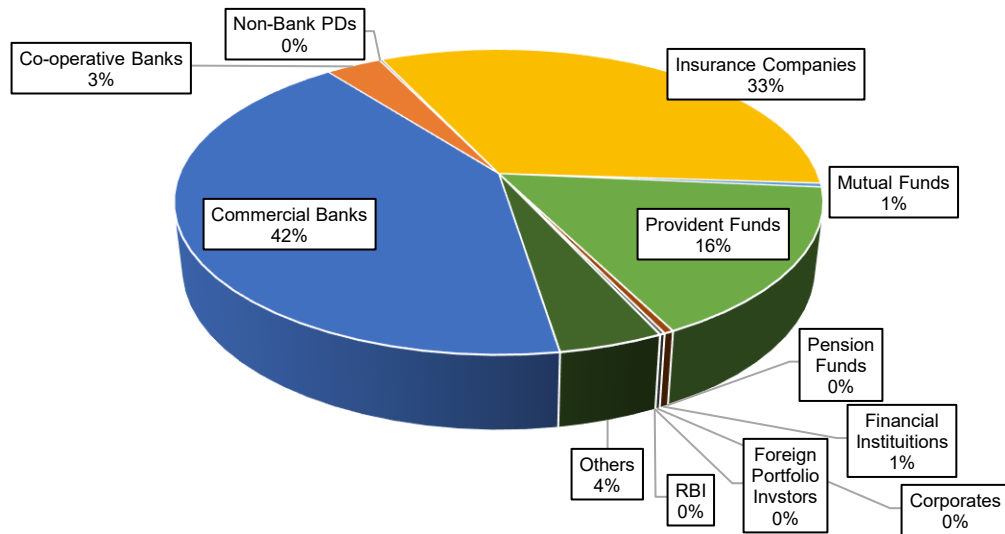
Central Government - End March 2007



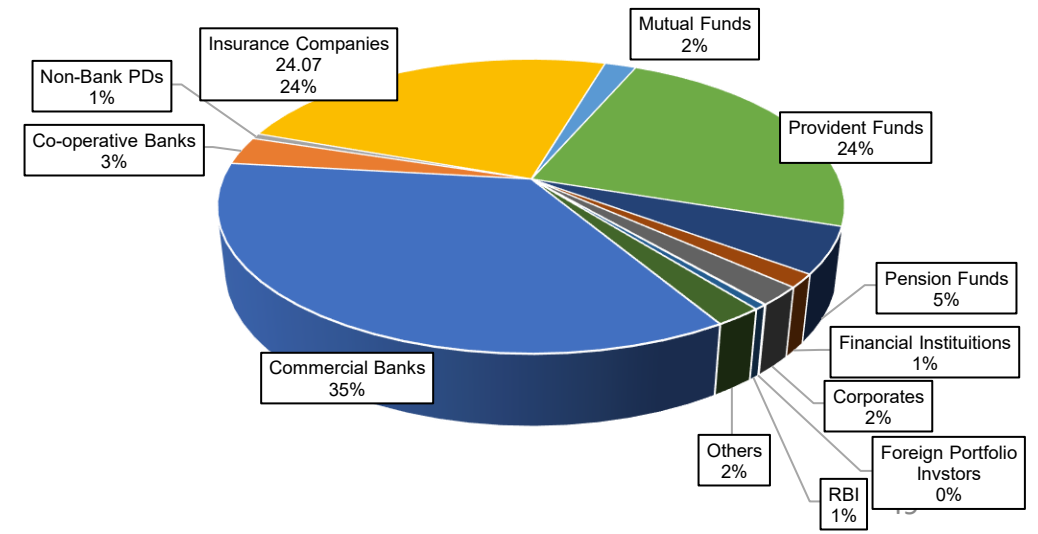
Central Government - End March 2025



State Government - End June 2015

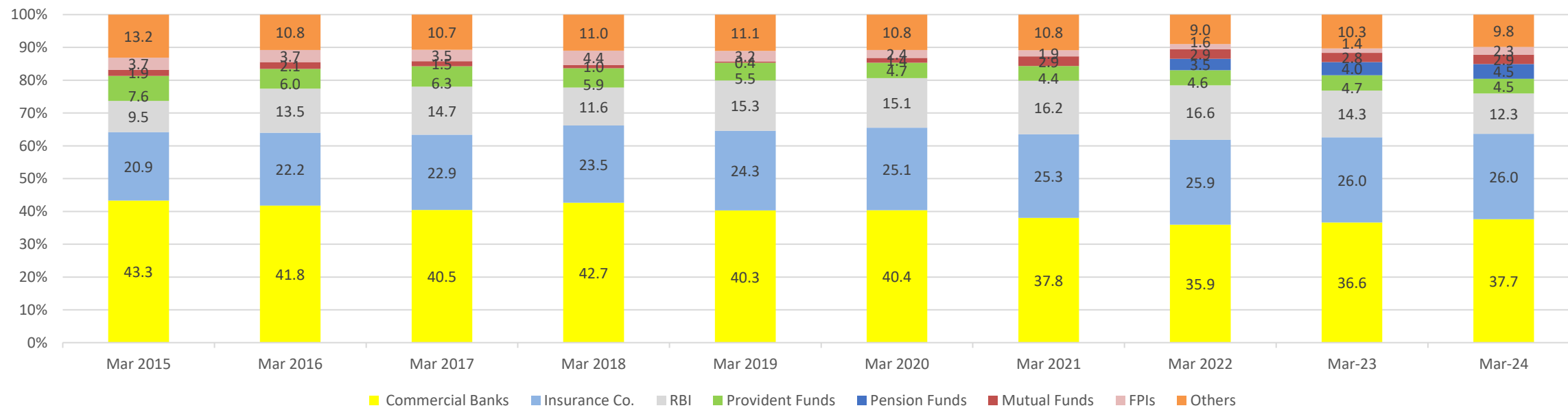


State Government - End March 2025



Diversification of Investor Base of G-sec market in India

50



- Commercial banks continue to have the largest share in G-sec markets (38%) though its share has declined in recent years
- Share of Insurance Companies has increased significantly over the years: 21% in 2015 to 26% as on March 2024
- Pension Funds have emerged as an important investor category and their share is now around 4.5%.
- Share of Provident Funds has remained around 4.5% over the last few years.
- RBI ownership has reduced in last two years.

Debt Management Framework

Cost Optimization

Demand Estimation

Instruments catering to
different investors class

Transparency – Issuance
calendar, publication of
public debt data

Risk Mitigation

Rollover Risk

Interest Rate Risk

Currency Risk

Market Development

Improving Liquidity:
Issuance of benchmark
securities on regular basis

Widening of Investor
base

Strengthening of Market
Infrastructure

Strategic Benchmark Indicators	Benchmark
Share of short-term debt in marketable debt stock	11±3 %
Weighted average maturity of outstanding debt	13±2 yr
Share of Inflation Indexed & floating rate debt in the issuances during a year	7±4 %
Share of external debt in public debt	7±3 %

G-sec Market Development through various Reforms

Automatic monetization deficit of Government of India's discontinued since 1997

Fiscal Consolidation through FRBM

- Government of India: FRBM Act 2003
- State Governments: All states have enacted their own FRBM Act, started with Karnataka (2002)

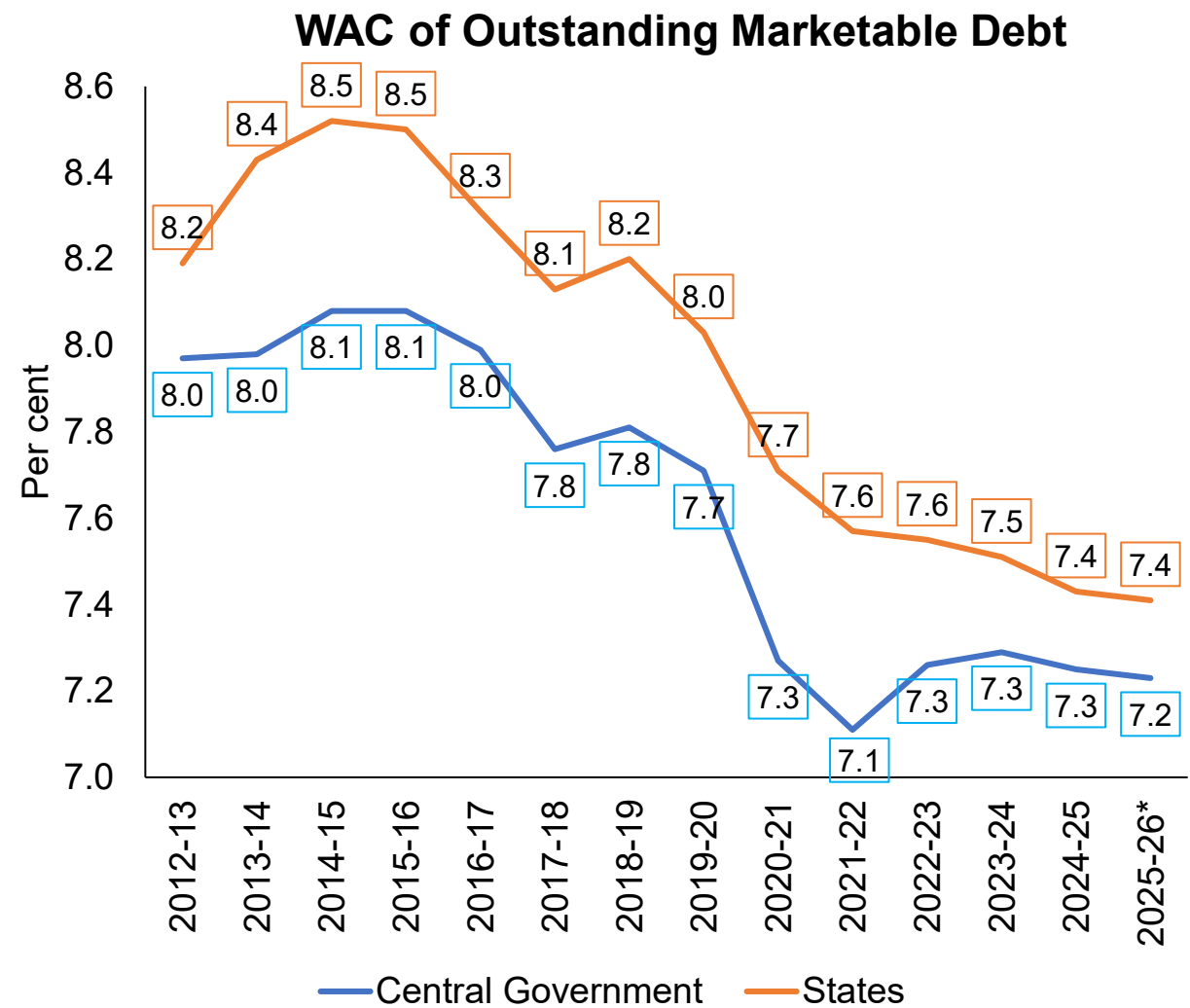
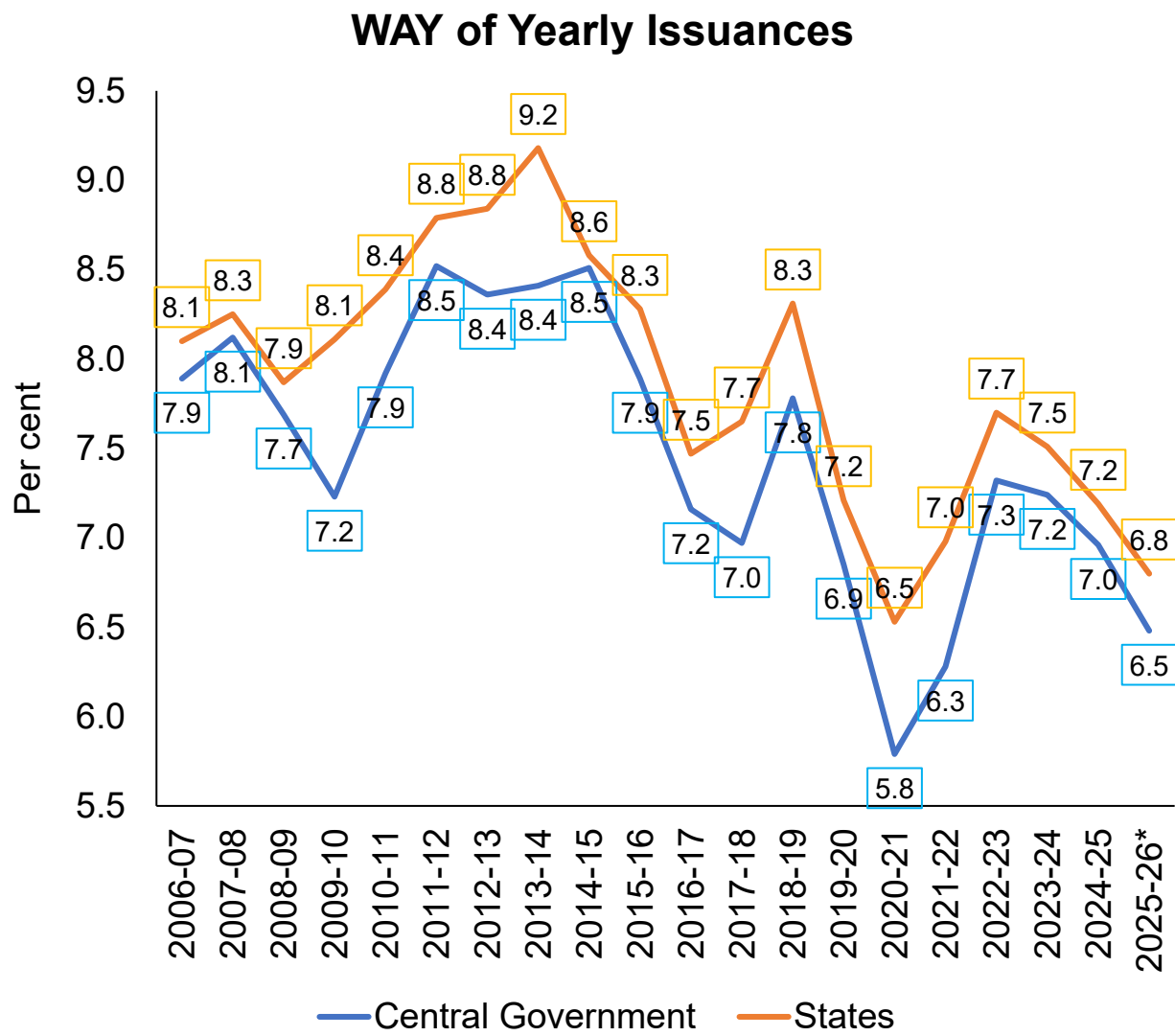
Market Borrowing: Government of India

- Auction method (1992-93)
- Half-yearly Indicative Calendar for issuance of G-sec (2002-03)
- Benchmark Issuance strategy (2020-21)
- Sovereign Green Bonds (2022-23)
- Introduction of new Benchmarks *viz.*, 7-year, 3-year security replaced 2-year, 50-years

Market Borrowing: State Governments

- Fully administered system
 - Combined borrowings of states conducted by Centre
 - SGS issued at pre-determined coupon rates , uniform for all the states
 - State's cost of borrowing aligned with Centre
- First auction for SGS in 1999 (Punjab, 60 crore, 10-year SGS)
- XII-FC allowed states to borrow directly from market

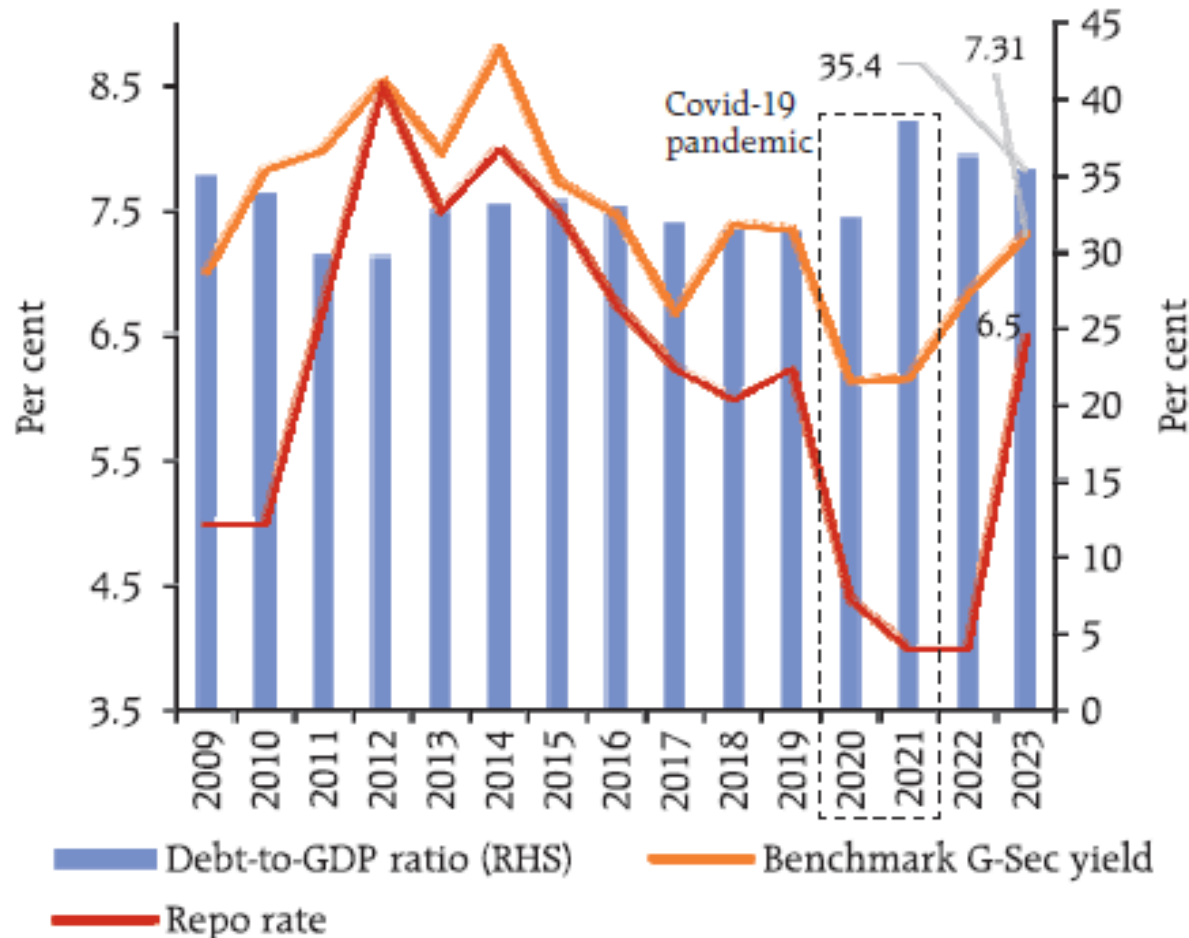
Cost of Market Borrowings



RBI attempted soft yield curve control during COVID-19 Pandemic

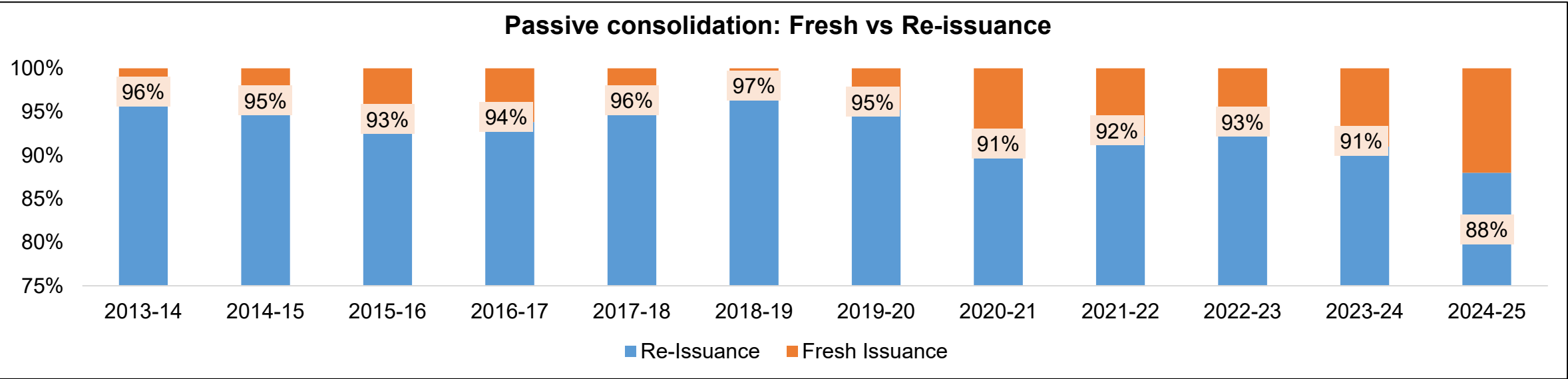
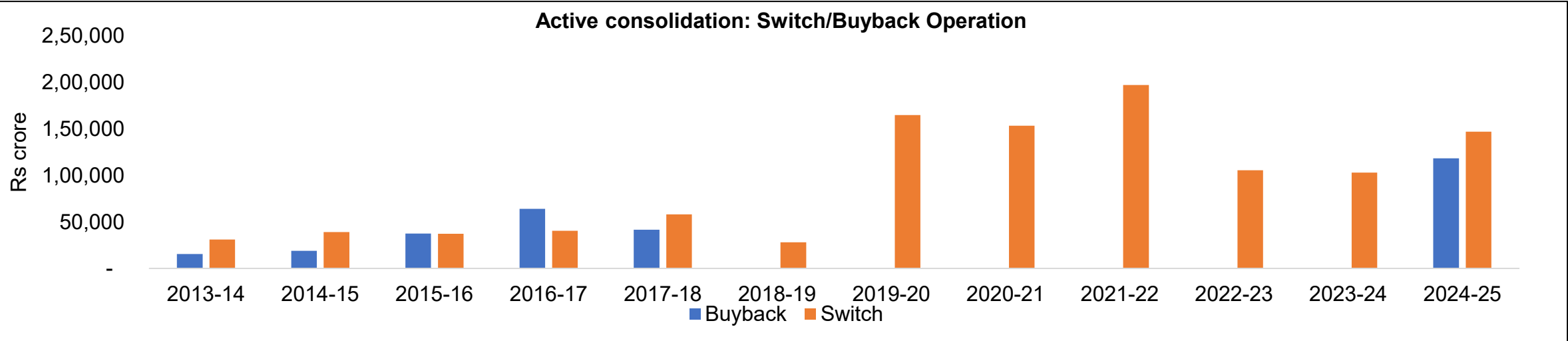
54

Movement in Benchmark G-sec Yield



- During COVID-19 RBI attempted to cut off yields in the multiple price auctions with a view to maintain 10-year yield at 6% or underneath.
- Auction left large unsubscribed portions
- Uniform price auction to prevent winners curse also did not help
- RBI had to abandon soft YCC & yields had to rise to clear the market

Central Government Debt Consolidation



Market Ecosystem

Primary market

- Transparent issuance process
 - Auctions since 1992 – electronic bidding platform
 - Auction system: Uniform/Multiple price auctions
 - Non-competitive bidding for retail investors
- **Wider bouquet of Government securities**
 - Dated securities, T-bills, Sovereign Green Bonds, STRIPs
- **Benchmark security issuance strategy**
 - Fixed coupon securities: 3, 5, 7, 10, 15, 30, 40, 50-year tenors
 - Issuance once every four weeks (pairing two securities each week)
- When-issued market
- Primary Dealer system – dated issuances are fully underwritten

Secondary market

- A separate electronic trading platform – NDS-OM
 - Liquidity pooled in a single venue
 - Anonymous order matching and Request For Quote (RFQ)
- Short-sale and Market Repo
- Near real time dissemination of trade information
- Central clearing of all trades on T+1/T+2
- Wider base of market participants
 - Banks, PDs, Mutual Funds, Insurance Companies, Pension Funds,
 - Foreign portfolio investors: multiple routes (General Route, VRR, FAR)

FPIs Investment in G-Secs

Route	Sept 2023	June 27, 2024	March 2025	July 31, 2025	Aug 04, 2025
MTF	71,105	73,544	66,496	56,097	57,602
VRR	1,083	2,148	802	1,005	1,005
FAR	96,661	1,84,762	3,06,249	2,84,535	2,83,025
Total	1,68,848	2,60,455	3,73,547	3,41,636	3,41,632

** Indian government bonds (IGBs) were included in GBI-EM global index suite starting June 28, 2024, and in the Bloomberg Emerging Market (EM) Local Currency Government Index from January 31, 2025.*

Price of the Bond is Inversely proportional to the Yield of the Bond

58

Inverse Relationship between Interest Rates and Bond Prices in the Bond Market

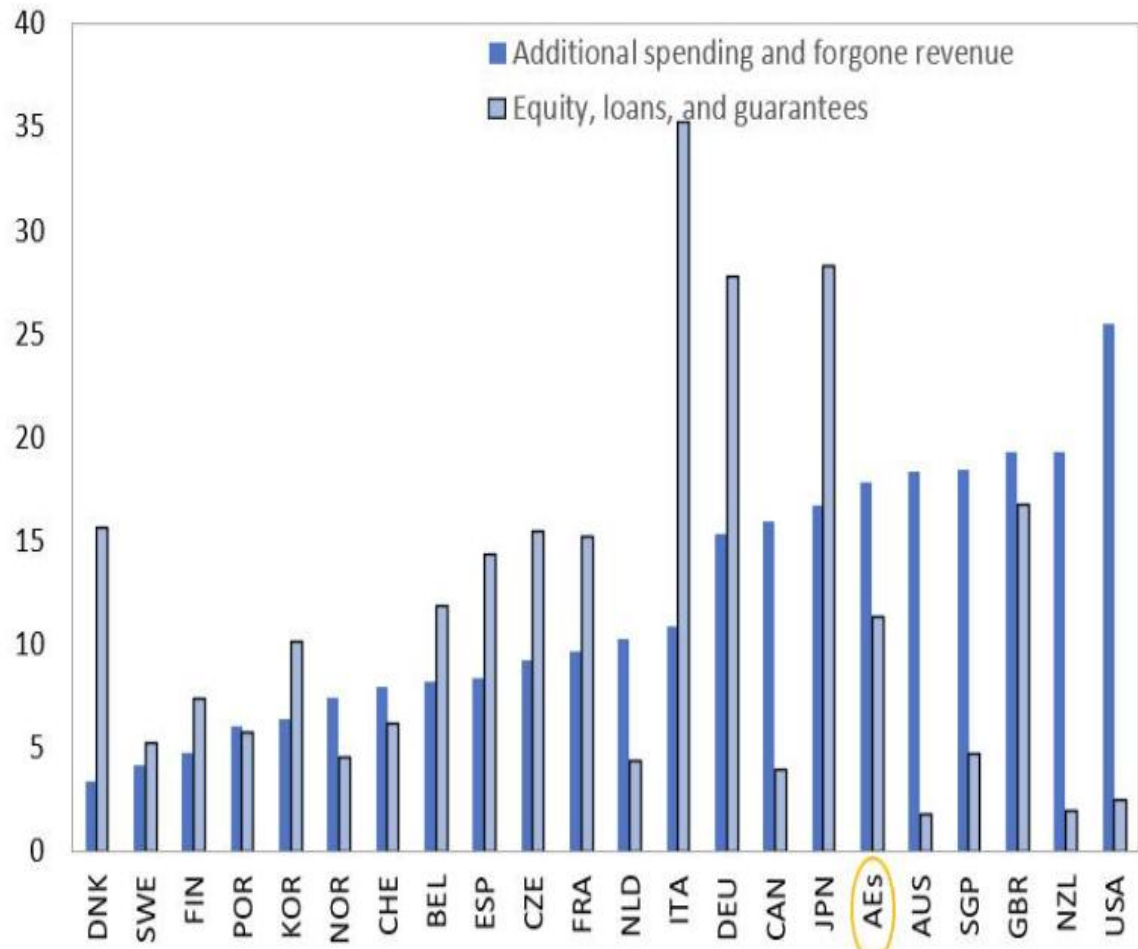
Interest rate trend	Effect on existing bond prices
	
	

- If market interest rate levels rise, the price of a bond falls. Conversely, if interest rates or market yields decline, the price of the bond rises. In other words, the yield of a bond is inversely related to its price.
- This is because to if yields have risen, a purchaser will buy previously issued bond only at discount to its price, while if yields have fallen, the existing holder of bond carrying higher yields will sell them only at a premium
- The relationship between yield to maturity and coupon rate of bond may be stated as follows:
 - When the market price of the bond is less than the face value, i.e., the bond sells at a discount, $YTM > \text{coupon yield}$.
 - When the market price of the bond is more than its face value, i.e., the bond sells at a premium, $\text{coupon yield} > YTM$.
 - When the market price of the bond is equal to its face value, i.e., the bond sells at par, $YTM = \text{coupon yield}$.

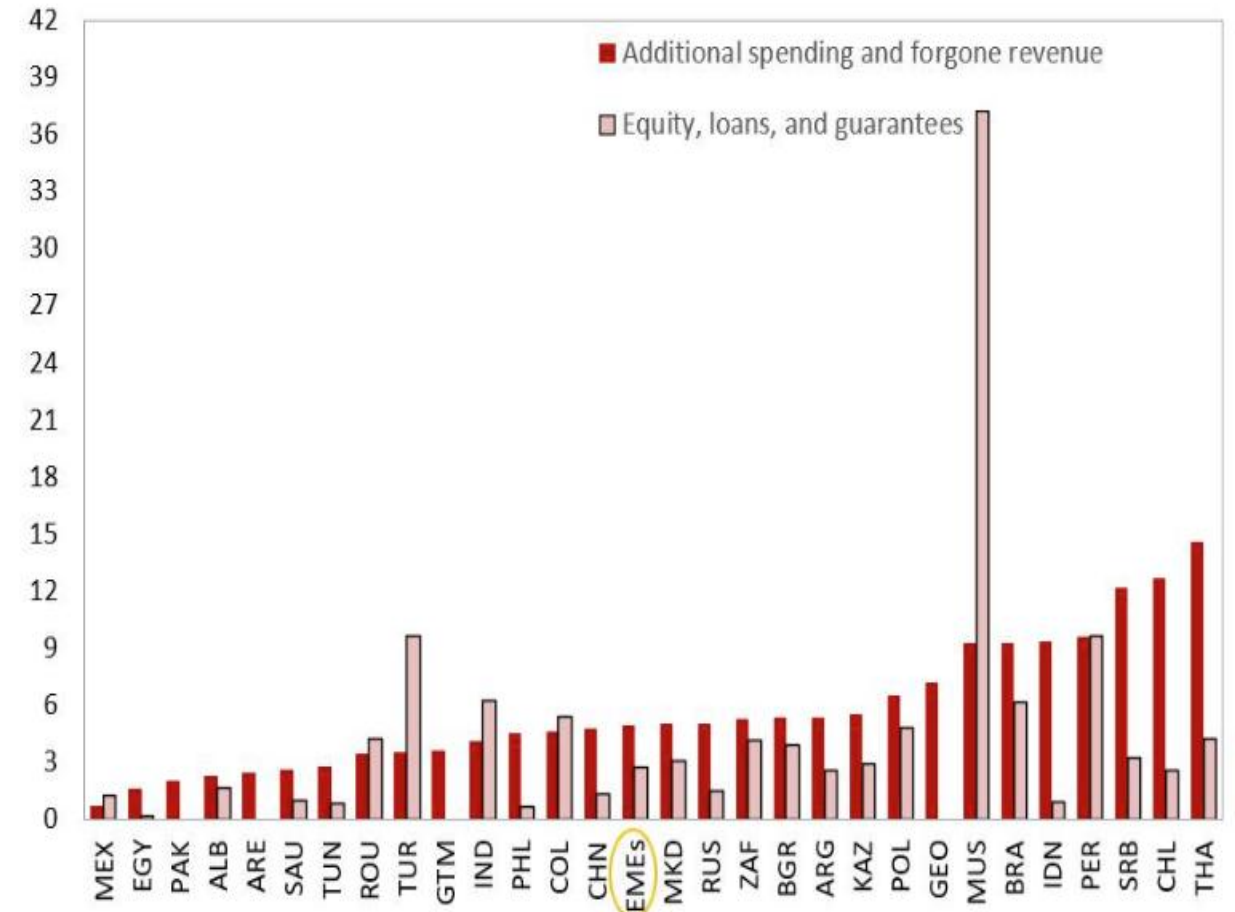
Unprecedented fiscal support provided during pandemic has widened fiscal gaps and debt sustainability in many countries

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ADVANCED ECONOMIES



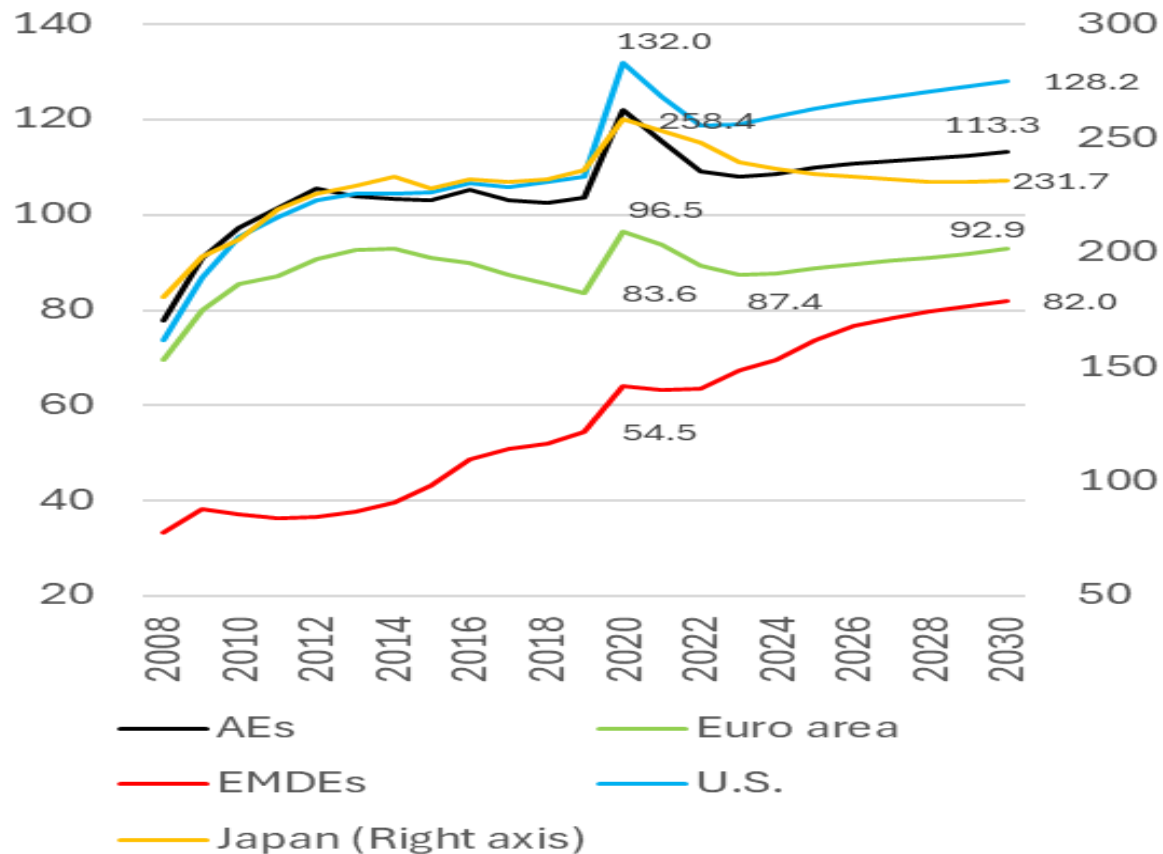
EMERGING MARKETS



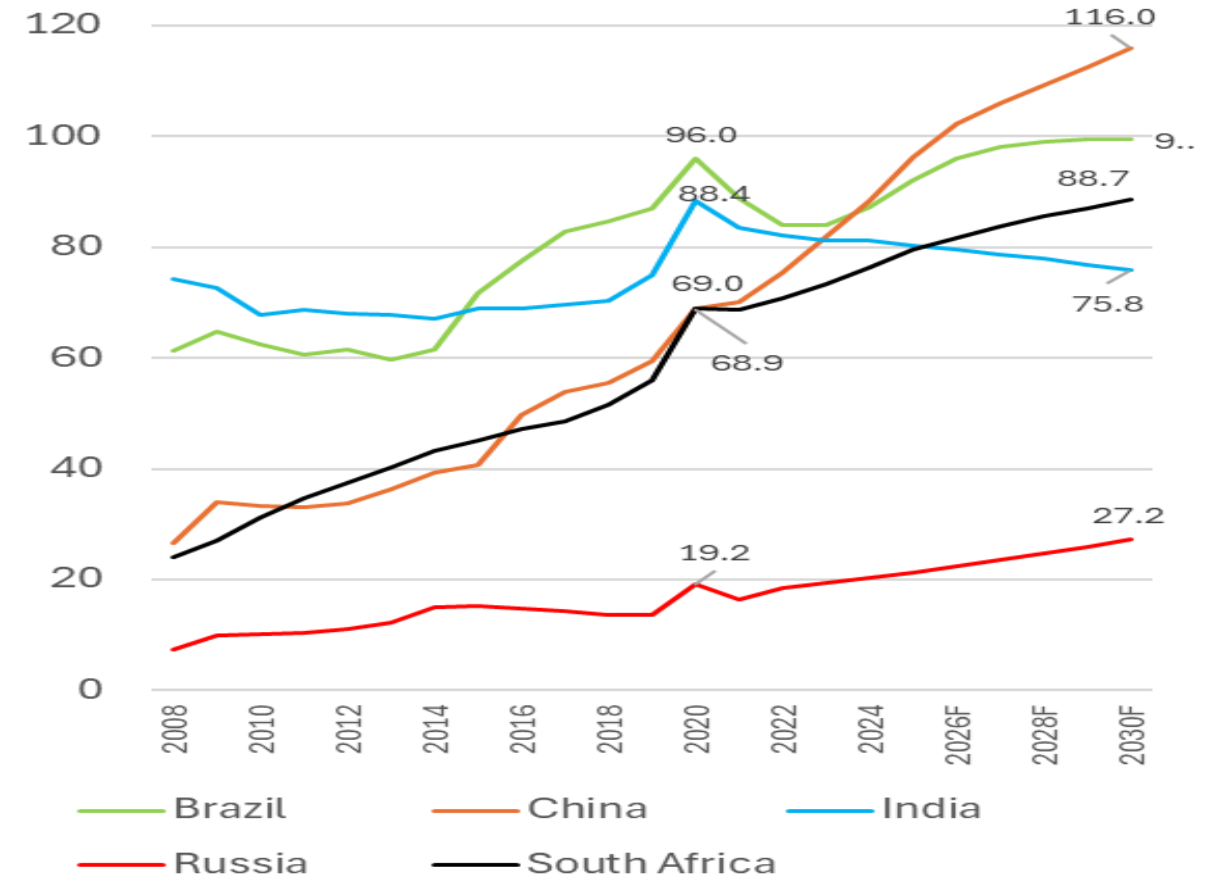
With global economy slowing down, fiscal space is limited; and fresh fiscal stimuluses could risk sovereign debt defaults

60

General government gross debt to GDP ratios in select country groups/ countries (%)

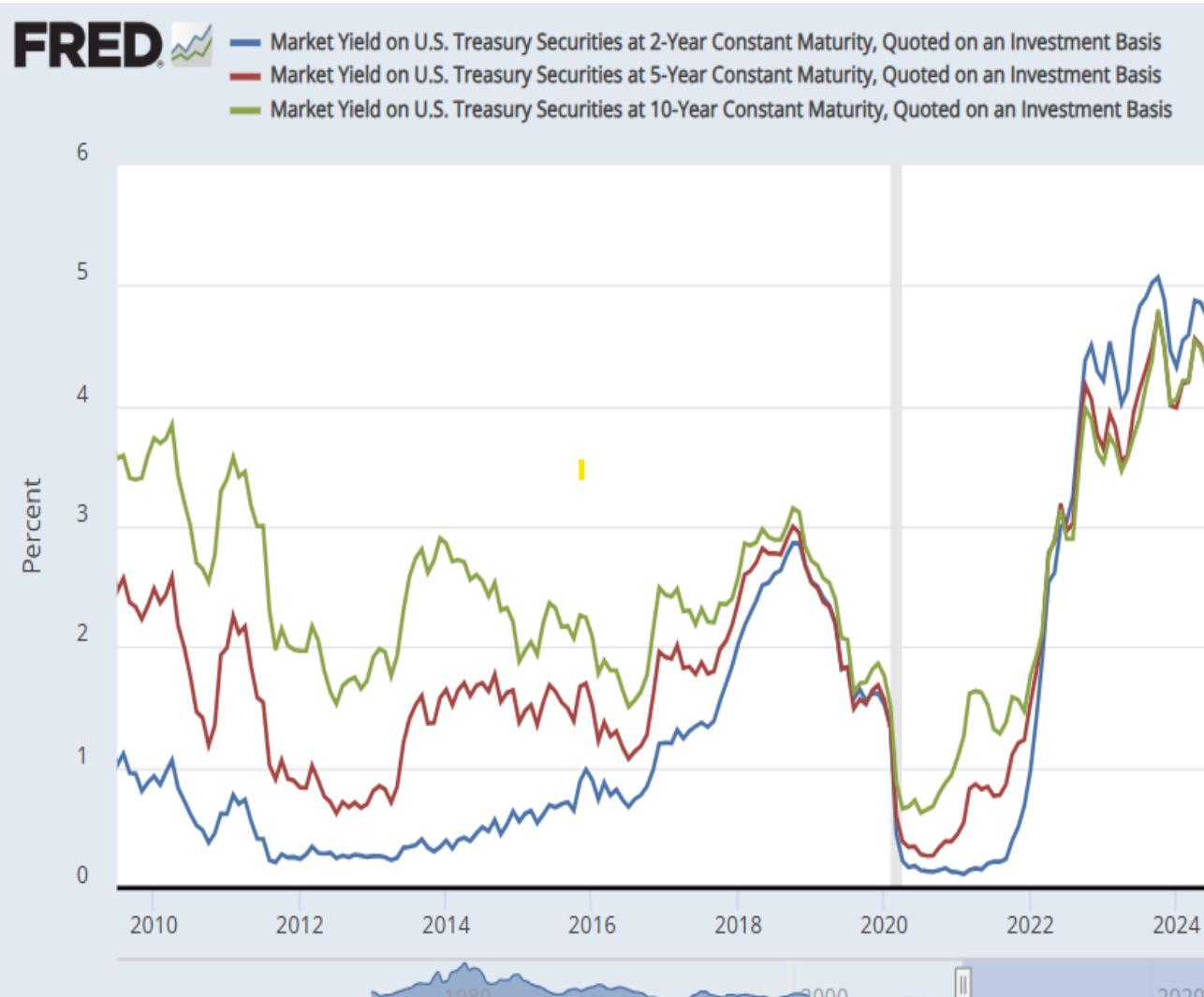


General government gross debt to GDP ratios in select EMDEs (%)



Yields of the Bond gyrate to global monetary policy cycles

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- Therefore, movements in global interest rate cycles, often started by Fed's monetary policy actions of raising or decreasing its policy rate, has a pronounced affect on bond yields (& prices).
- For instance, when pandemic struck, bond yields and term premia collapsed as investors factored in current and expected Fed actions of cutting target Federal funds rate to zero.
- However, when inflation spiked after pandemic receded, bond yields surged in response to current and expected monetary tightening

Primary Dealer System in Nutshell

- PDs system was introduced in 1995 to strengthen the G-Sec market infrastructure.
- PD system expanded and banks were permitted to act as a PD departmentally from 2006-07.
- At present there are 7 Standalone Primary Dealers (SPD) and 14 Bank Primary Dealers (Bank PD).

Role in Primary Market	Role in Secondary Market
▪ Participation in underwriting and dated securities primary auctions. ▪ To participate in T-Bills Auction ▪ Provide feedback on assessment of market demand that aids in formulation of the half-yearly issuance calendar for Central Government securities.	Play an important role participating in OMO, When-issued, short selling, stripping of G-secs, market making in G-secs, etc. To achieve certain minimum secondary market turnover. Participate in Retail Direct Scheme – Market Making – presence on NDS-OM (Odd lot and RFQ).

Facilities provided to PDs:

Remuneration by payment of Underwriting Commission

Current account with RBI, e-Kuber access, SGL facility, membership of trading platforms.

Permission to borrow and lend in money market

Access to Liquidity Adjustment Facility, OMOs

Standing Liquidity Facility (SLF) for Standalone Primary Dealers.

More on Primary Dealer (PD) System for G-sec market in India

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- The system of Primary Dealers (PDs) in the G-sec market was introduced by RBI in 1995 (& operational from 1996) to strengthen the market infrastructure and was aimed at not only a more efficient primary market but also to put in place an improved, efficient secondary market trading system in G-secs
- Initially RBI set up Discount and Finance House of India (DFHI) and Securities Trading Corporation of India (STCI) as the two initial PDs and off-loaded them later. As of end-March 2024, there were 21 PDs of which 7 were stand-alone while 14 were bank-PDs.
- PDs functioning include: (i) dealing and underwriting in G-secs, (ii) dealing in Interest Rate Derivatives, (iii) providing broking services in Government securities, (iv) dealing and underwriting in Corporate / PSU / FI bonds/ debentures, (v) lending in Call/ Notice/ Term/ Repo/ TREPS market, (vi) investment in CPs & CDs, (vii) investment in Security Receipts issued by Securitization Companies/ Reconstruction Companies, Asset Backed Securities (ABS), Mortgage Backed Securities (MBS).
- PDs ensure 100% of underwriting of a primary G-sec auction and in return are provided an incentive structure in form of underwriting commission. The underwriting commitment is divided into two parts, viz; (i) minimum underwriting commitment (MUC), and (ii) additional competitive underwriting (ACU). All successful bidders in the ACU get commission as per the auction rules.
- Bank-PDs are required to maintain separate books of accounts for transactions relating to PD business.

Widening of Instrument Choice in G-sec Market in India

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- G-secs today are catering to needs of diverse investor classes with long-tenors being demanded by insurance & pension funds
 - Dated securities: 3-yr to 50-year tenors
 - T-Bills: 91/182/364 days
 - Cash Management Bills (up to 91 days) (short-term instruments issued by Central government for temporary liquidity mismatches)
 - State Government Securities (SDLs)
 - STRIPS (strip is a bond coupon that has been removed from the bond so that the two parts can be sold separately, as an interest-paying coupon bond and as a zero-coupon bond).
 - Sovereign Gold Bonds (SGBs) with underlying as Gold price
 - Sovereign Green Bonds
 - **Benchmark security issuance strategy**
 - Fixed coupon securities with tenors 3, 5, 7, 10, 15, 30, 40, 50-year are issued at regular interval
 - 7-yr benchmark was introduced in April 2022; 3-yr benchmark in April 2023; 50-yr security introduced in H2:2023-24
- RBI publishes calendar of issuance is published on half yearly basis for information of investors

Some Treasury Securities & their features

65

SLR (Gov Securities/ State Government) Securities

Fixed Coupon Bonds
Floating Rate Bonds (FRBs)
Sovereign Green Bonds (SGrBs)
Tenors upto 50 yrs

**G-sec/
SDLs**

T-Bills

Treasury Bills

Money market instrument issued at discount;
91-d T-bill, 182-d T-bill & 364-d T-bill

Floating Rate Savings Bond

Issued for min amount of ₹ 1000 and multiples thereof
Coupon is 35 bps above NSC rate
Term is 7 years
Pre-mature redemption only for senior citizens

**FRSB,
2020**

SGB

Sovereign Gold Bond

- Price linked to market price of Gold; discount of Rs 50 /gm on nominal value through digital mode
- Min. investment of 1 gm & max. of 4 kg/year for individuals and HUF;
- Maturity of 8 yrs, pre-mature redemption after 5th year of issuance date;
- Interest rate of 2.5% per annum on the nominal value payable on half yearly

Transparency in Borrowing Process

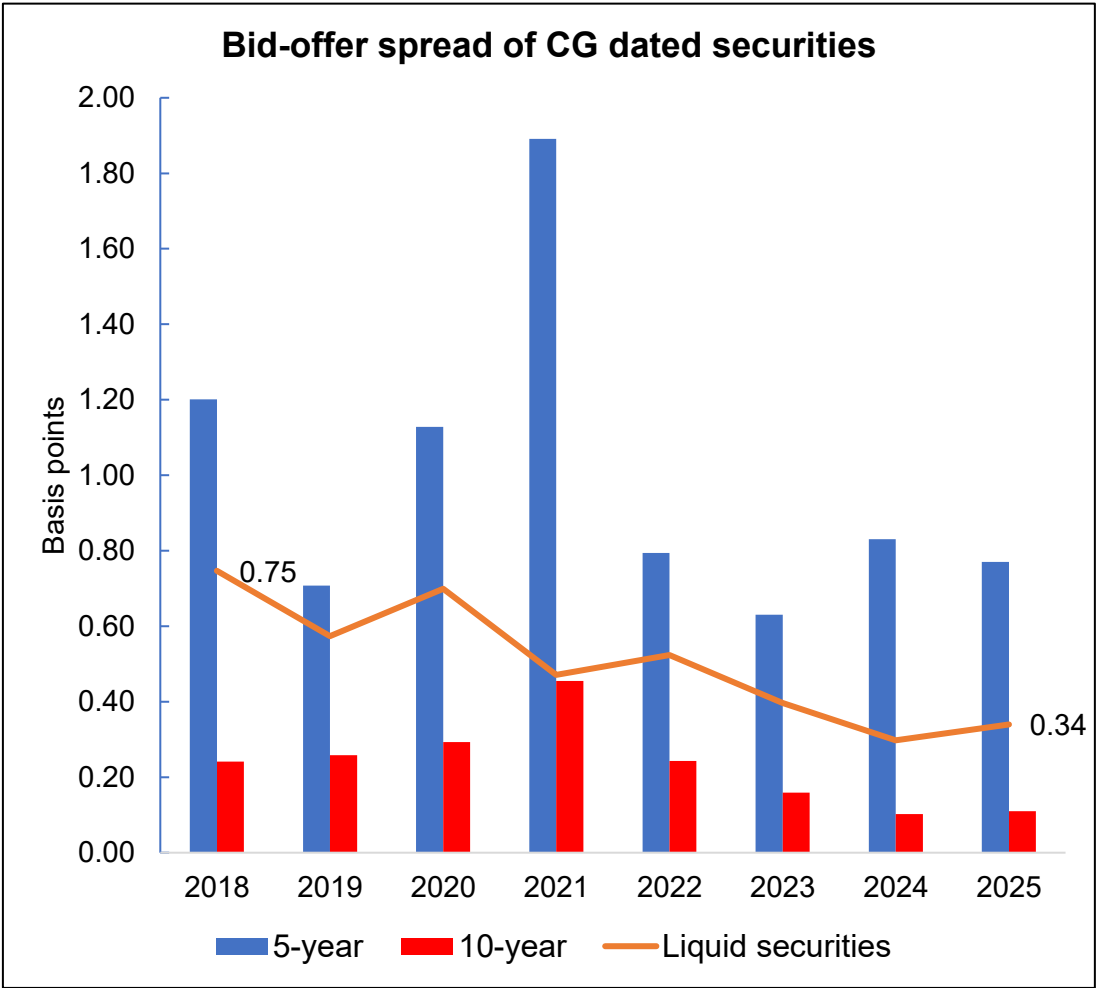
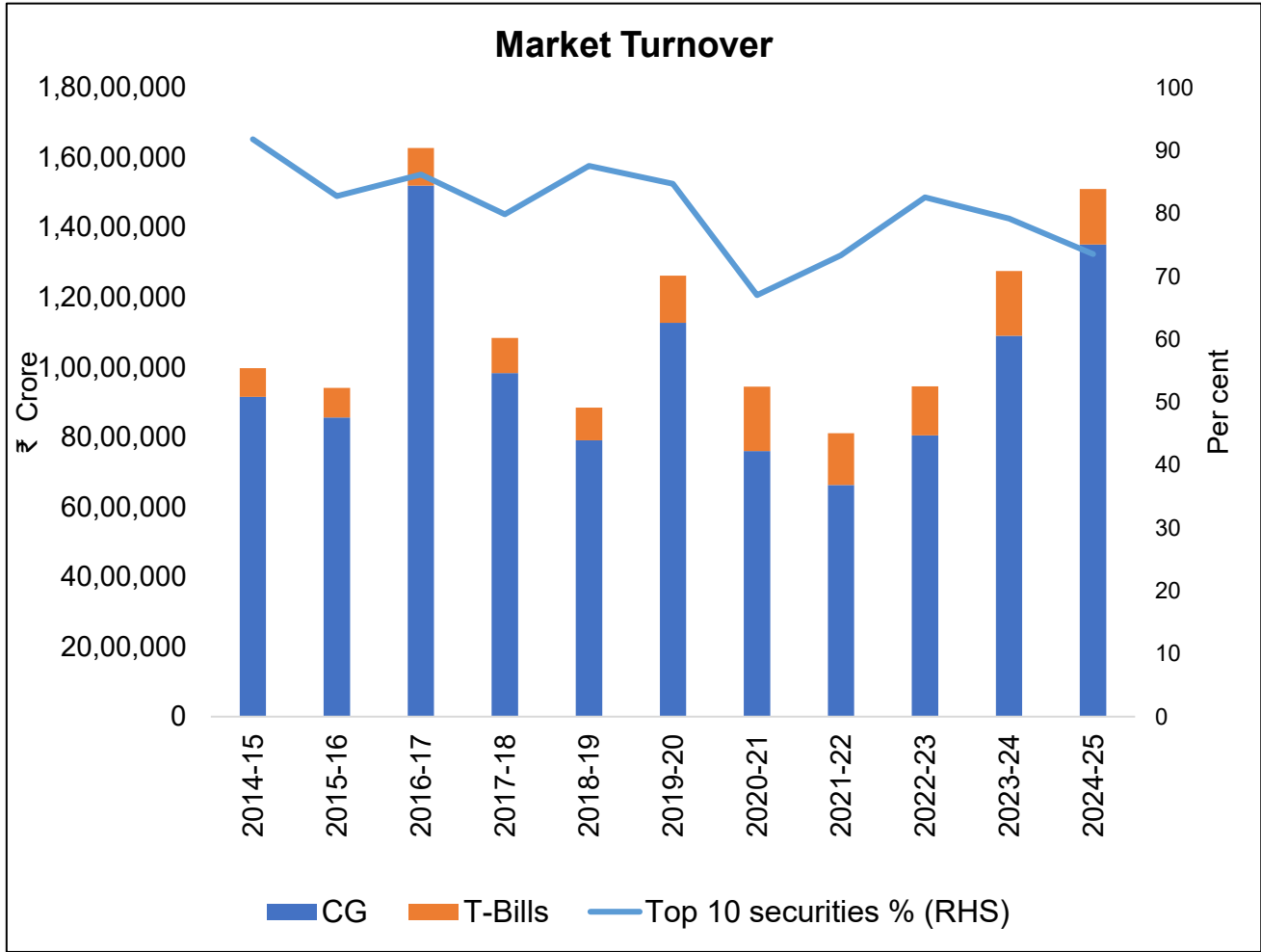
Quarterly/Half Year Calendars and Pre-Post Auction Press Releases

- Auction Calendar for instruments (G-sec/SGS and T-bills) are released on the RBI website, which convey to the market participants on the amount and the name of the issuers (SGS) likely to float their papers
- Before conduct of weekly auction, the market is informed of the ensuing auction amount through Press Release
- Press Release on the cut-off yield, tenors and notified and accepted amount is also released on the RBI website on the day of auction

Dissemination of Public Debt Statistics

- RBI releases time series data on outstanding liabilities of G-sec and SGS
- RBI also releases data on STRIPs in G-secs
- RBI Bulletin releases data on Market Borrowings by SGS, along with the investment position in CSF/GRF and ATBs of States

Market Liquidity



Are G-secs Risk free? If not, what are the risks... (1)

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- G-Secs are generally referred to as risk free instruments as sovereigns rarely default on their payments. However, as is the case with any financial instrument, there are risks associated with holding the G-Secs. Hence, it is important to identify and understand such risks and take appropriate measures for mitigation of the same. The following are the major risks associated with holding G-Secs:
- **Market risk:** Arises out of adverse movement of prices of the securities due to changes in interest rates. This will result in valuation losses on marking to market or realizing a loss if the securities are sold at adverse prices. Small investors, to some extent, can mitigate market risk by holding the bonds till maturity so that they can realize the yield at which the securities were actually bought.
- **Reinvestment risk:** Cash flows on a G-Sec includes a coupon every half year and repayment of principal at maturity. These cash flows need to be reinvested whenever they are paid. Hence there is a risk that the investor may not be able to reinvest these proceeds at yield prevalent at the time of making investment due to decrease in interest rates prevailing at the time of receipt of cash flows by investors.

Are G-secs Risk free? If not, what are the risks...(2)

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- **Liquidity risk:** Liquidity in G-Secs is referred to as the ease with which security can be bought and sold i.e. availability of buy-sell quotes with narrow spreads. Liquidity risk refers to the inability of an investor to liquidate (sell) his holdings due to non-availability of buyers for the security, i.e., no trading activity in that particular security or circumstances resulting in distressed sale (selling at a much lower price than its holding cost) causing loss to the seller. Usually, when a liquid bond of fixed maturity is bought, its tenor gets reduced due to time decay. For example, a 10-year security will become 8-year security after 2 years due to which it may become illiquid.
- The bonds also become illiquid when there are no frequent reissuances by the issuer (RBI) in those bonds. Bonds are generally reissued till a sizeable amount becomes outstanding under that bond. However, issuer and sovereign have to ensure that there is no excess burden on Government at the time of maturity of the bond as very large amount maturing on a single day may affect the fiscal position of Government. Hence, reissuances for securities are generally stopped after outstanding under that bond touches a particular limit. Due to illiquidity, the investor may need to sell at adverse prices in case of urgent funds requirement. However, in such cases, eligible investors can participate in market repo and borrow the money against the collateral of such securities.

Comparison of US and Indian Government Securities Market

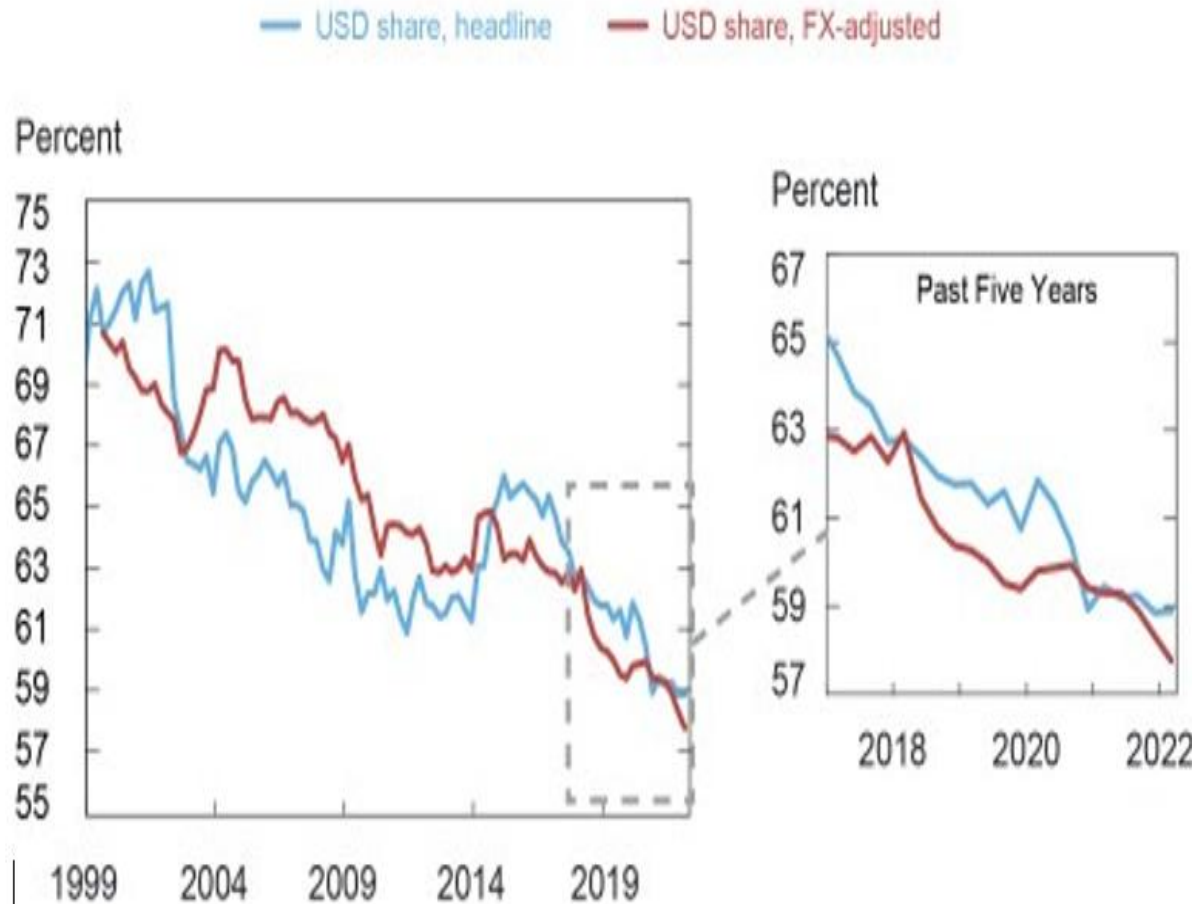
71

	India	USA
Investor base	Domestically held. Dominated by banks, followed by insurance, limited foreign and retail participation	Broad based. Include domestic institution, individuals, foreign central banks, mutual funds and pension funds
Retail Participation	Low retail participation	Significant Retail participation
Foreign Investment	Foreign Portfolio Investors (FPIs) have limited share due to investment caps and currency risk.	Foreign investors (esp. central banks like China, Japan) are major holders
Market Infrastructure	NDS-OM	OTC
Liquidity	Concentrated in few benchmarks	Highly liquid

FX market is rapidly changing in microstructure

72

US Dollar Share in Global Reserves has fallen over last 25 years



- What is the structure of the FX global market? What are the instruments
- Some recent developments in global FX markets –
 - (i) Global FX Code;
 - (ii) high-frequency trading (HFT) and fast-paced electronic platforms
- Central bank interventions in FX markets
- Dollar shortages and swap lines
- Dollar-dominance & dollar de-dollarization

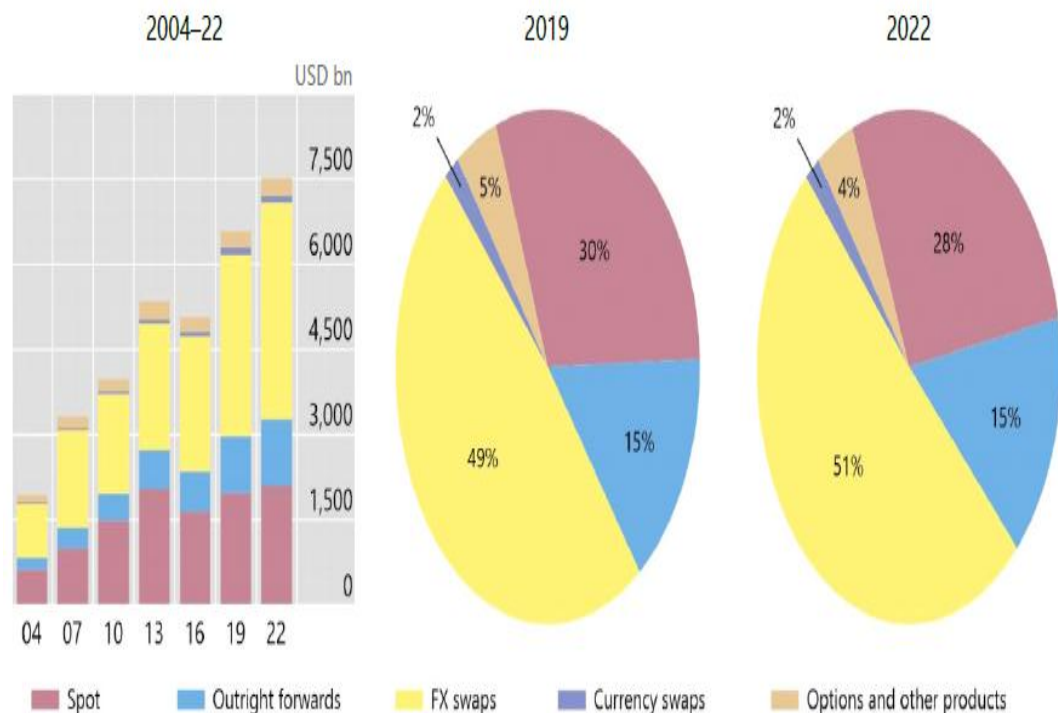
Global FX Markets are predominantly OTC Markets, though fast-paced Electronic Markets are starting to gain popularity

73

Foreign exchange market turnover by instrument¹

Net-net basis, daily averages in April

Graph 1



¹ Adjusted for local and cross-border inter-dealer double-counting, ie "net-net" basis.

- Daily OTC FX markets trading touched \$7.5 trillion in April 2022 ("net-net" basis, all FX instruments), up 14% from \$6.6 tn in 2019.
- Turnover of FX swaps accounted for 51% of global turnover, up from 49% in 2019. The share of spot trades fell to 28% from 30% in 2019, and that of outright forwards remained unchanged at 15%.
- Overall, inter-dealer trading, ie trading among "reporting dealers", reached \$3.5 trillion or 46% of global turnover in April 2022, a higher share than in previous Surveys. Trading with "other financial institutions" saw modest growth and accounted for 48% of global turnover (down from 55% in 2019).
- The US dollar was on one side of 88% of all trades (unchanged from 2019). The share for the euro decreased marginally to 31% (from 32% in 2019), and those for the Japanese yen and the pound sterling remained unchanged at 17% and 13%, respectively. The renminbi's share rose to 7%, making it the fifth most traded currency in 2022 (up from eighth place in 2019 with a 4% share).
- Trading at sales desks in five jurisdictions – the UK, the US, Hong Kong SAR, Singapore and Japan – amounted to 78% of all FX trading ("net-gross" basis). Trading activity in the United States and Singapore grew by more than the global average.

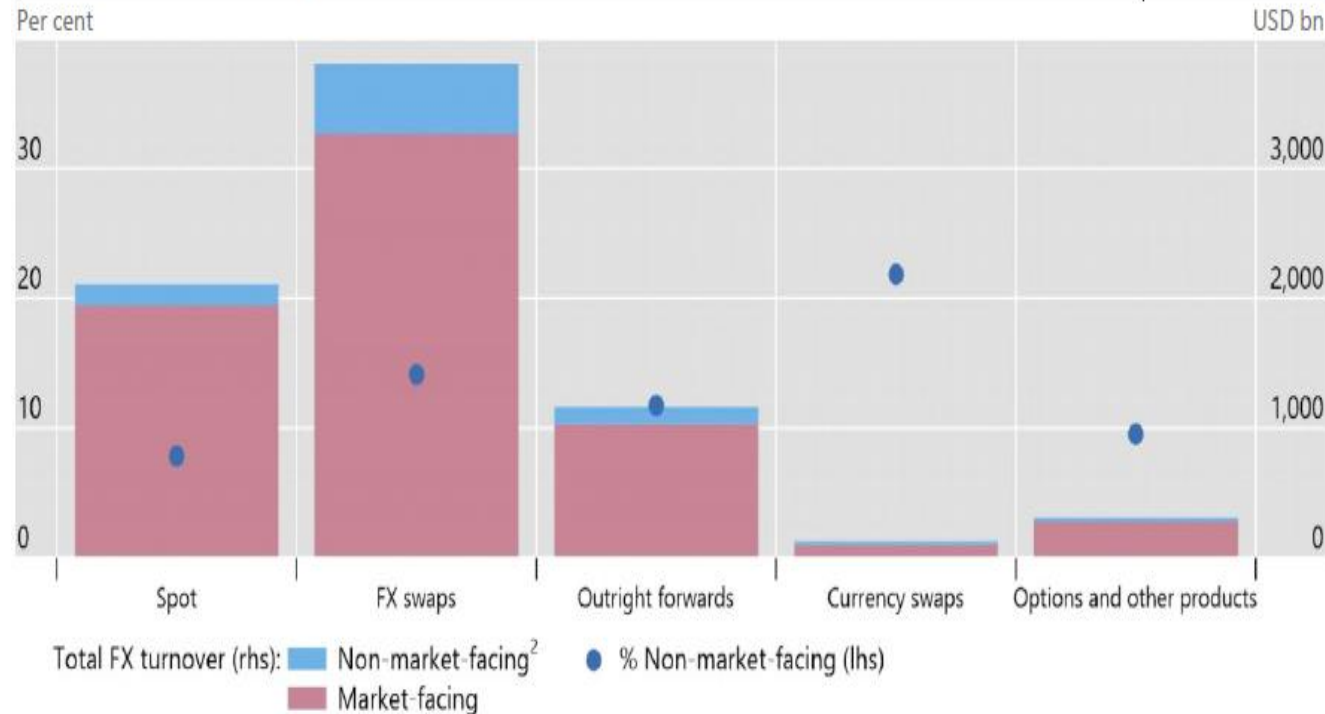
FX swaps & spot trades most popular in FX markets; together they accounted for 79% of the turnover in 2022

74

Total foreign exchange turnover, by instrument and type of trade

Net-net basis,¹ daily average in April, in billions of US dollars

Graph 2



¹ Adjusted for local and cross-border inter-dealer double-counting. ² Back-to-back and compression trades.

Source: BIS Triennial Central Bank Survey.

- “Market-facing trades” are deals with customers and other unrelated entities that contribute to price formation in the market.
- “Non-market-facing” trades consist of (i) “back-to-back” trades, which are deals that automatically follow trades with customers to shift risk across sales desks; and (ii) compression trades, in which dealers optimize their portfolios by replacing existing contracts with new ones to reduce notional amounts while keeping net exposures unchanged.
- Total non-market-facing trades amounted to US\$895 billion or 12% of the \$7.5 trillion of global FX turnover in 2022

OTC FX turnover of US\$7.5tn is dominated by FX swaps & spot transactions

75

OTC foreign exchange turnover by instrument

"Net-net" basis, 1 daily averages in April in billions of US dollars Table 1

Instrument	2010	2013	2016	2019*	2022
Foreign exchange instruments	3,973	5,357	5,066	6,581	7,508
Spot transactions	1,489	2,047	1,652	1,979	2,107
Outright forwards	475	679	700	998	1,163
Foreign exchange swaps	1,759	2,240	2,378	3,198	3,810
Currency swaps	43	54	82	108	124
Options and other products ²	207	337	254	298	304
<i>Memo:</i>					
Turnover at April 2022 exchange rates ³	3,542	4,727	4,851	6,446	7,508
Exchange-traded derivatives ⁴	144	145	115	127	152

¹ Adjusted for local and cross-border inter-dealer double-counting (ie "net-net" basis). ² The category "other FX products" covers highly leveraged transactions and/or trades whose notional amount is variable and where a decomposition into individual plain vanilla components was impractical or impossible. ³ Non-US dollar legs of foreign currency transactions were converted into original currency amounts at average exchange rates for April of each survey year and then reconverted into US dollar amounts at average April 2022 exchange rates.

⁴ Euromoney Tradedata; Futures Industry Association; The Options Clearing Corporation; BIS derivatives statistics. Foreign exchange futures and options traded worldwide. * Revised data.

- FX turnovers in nominal terms have increased by ~2X (i.e doubled) in last 12 years
- FX swaps have more than doubled in turnover, while currency swaps have risen ~3X (i.e. trebled)

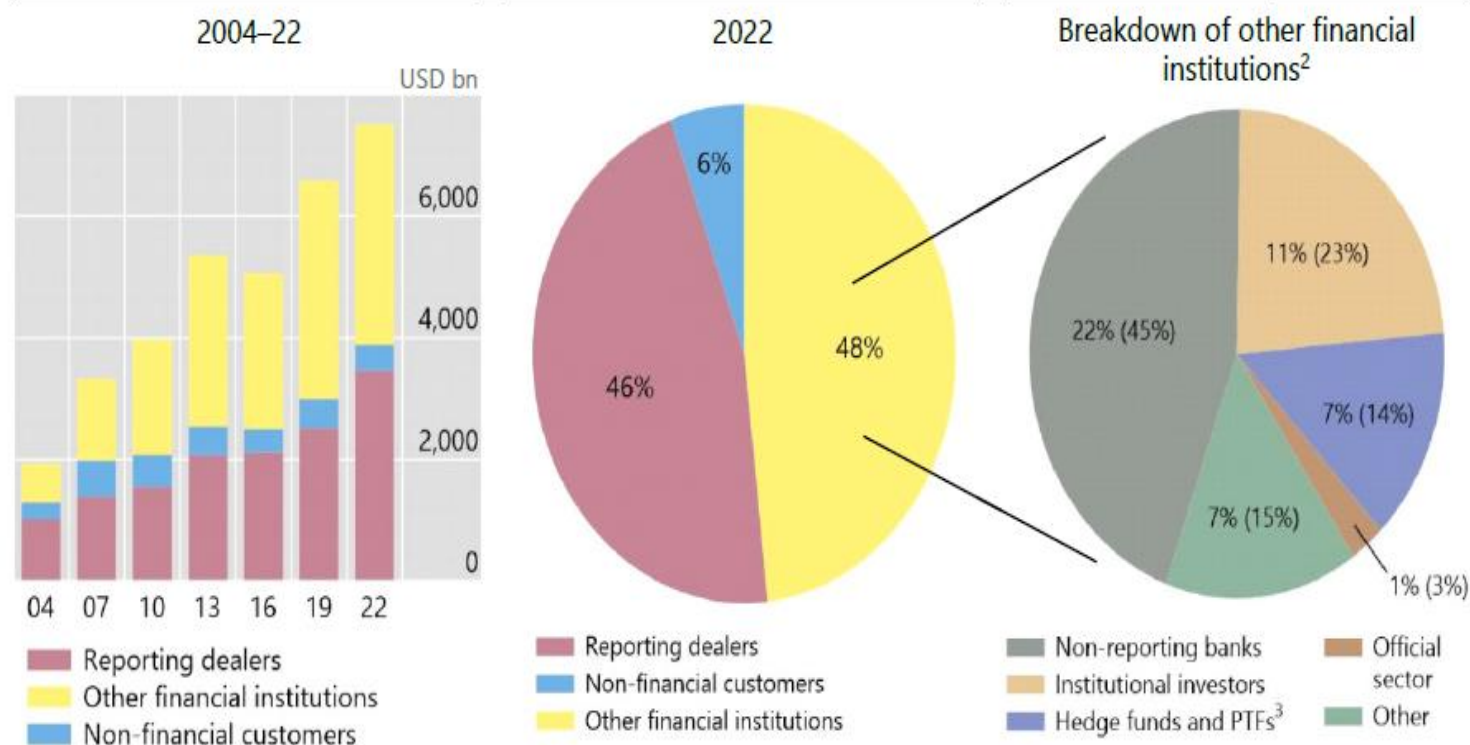
OFI and (reporting) FX dealers account for 94% of the FX market turnover

76

Foreign exchange market turnover by counterparty¹

Net-net basis, daily averages in April

Graph 3



- Reporting dealers are generally reporting banks
- Other financial institutions (OFIs) are non-reporting banks, hedge funds, proprietary trading firms (PTFs), institutional investors and official sector financial institutions

¹ Adjusted for local and cross-border inter-dealer double-counting, ie "net-net" basis. ² For definitions of counterparties, see explanatory notes in the Annex. Shares in overall total, with shares in other financial institutions in brackets. ³ Proprietary trading firms.

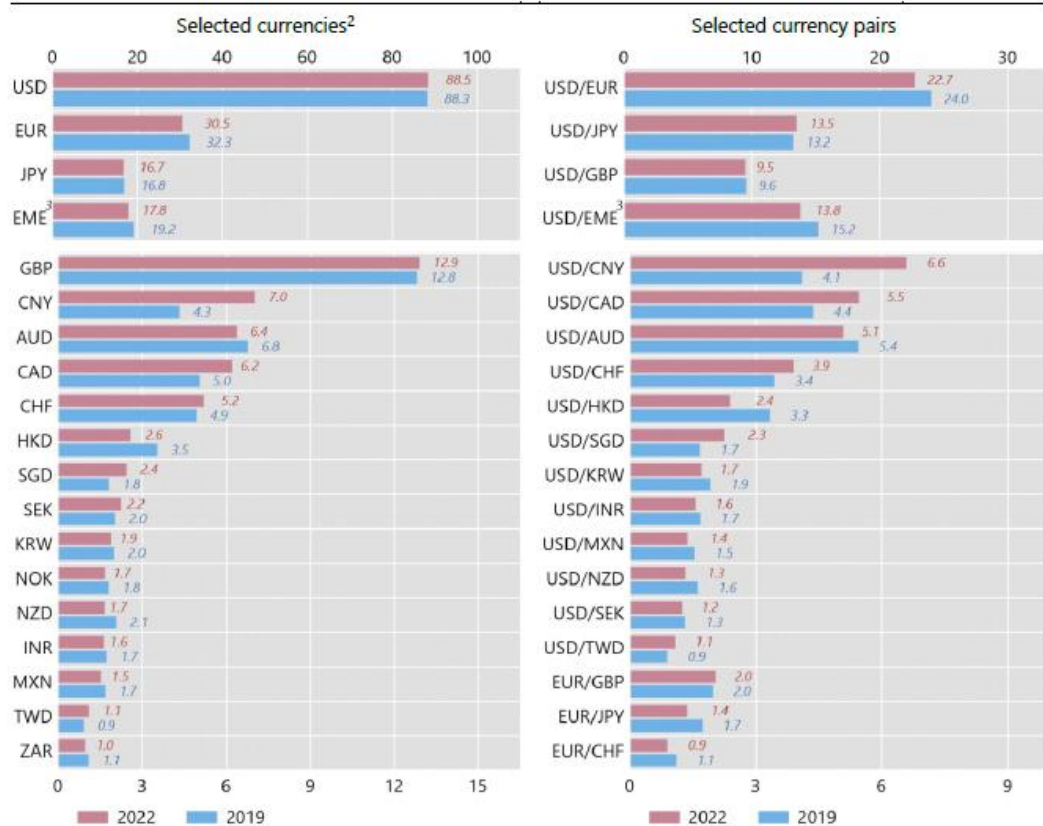
USD accounts for over 88% of the FX market turnover, with two currency pairs - USD/EUR & USD/JPY – together accounting for over 36% of FX turnover

77

Foreign exchange market turnover by currency and currency pairs¹

Net-net basis, daily averages in April, as a percentage of total turnover

Graph 4



¹ Adjusted for local and cross-border inter-dealer double-counting, ie "net-net" basis. ² As two currencies are involved in each transaction, the sum of shares in individual currencies will total 200%. ³ Emerging market economy currencies excluding the Chinese renminbi and Russian rouble: AED, ARS, BGN, BHD, BRL, CLP, COP, CZK, HKD, HUF, IDR, ILS, INR, KRW, MXN, MYR, PEN, PHP, PLN, RON, SAR, SGD, THB, TRY, TWD and ZAR.

Source: BIS Triennial Central Bank Survey. For additional data by currency and currency pairs, see Tables 4 and 5. See our [Statistics Explorer](#) for access to the full set of published data.

- US dollar (USD) remains the dominant currency in global FX markets
- CNY (also known as RMB) has sharply increased its share in FX market turnover from 4.3% in 2019 to 7.0% in 2022.
- This follows the inclusion of RMB in the SDR basket by the IMF at end-Sept 2016 (Two criteria for inclusion: (i) export criteria; (ii) freely usable criteria)
- However, the dominance of USD in international FX markets is unlikely to go away for a long time, even as most central banks are diversifying their FX reserves holdings

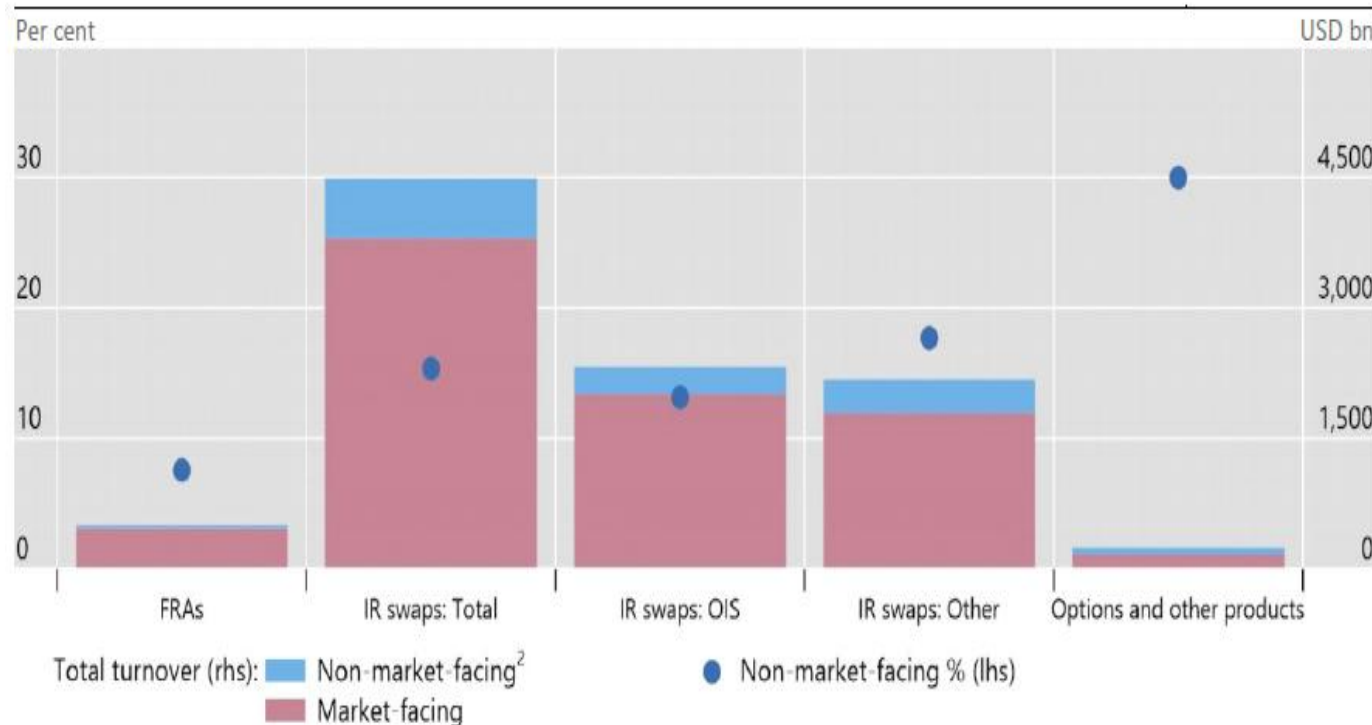
OIS swaps the largest component of the interest rate derivatives market

78

Total interest rate derivatives turnover, by instrument and type of trade

Net-net basis,¹ daily average in April, in billions of US dollars

Graph 2



¹ Adjusted for local and cross-border inter-dealer double-counting. ² Back-to-back and compression trades.

Source: BIS Triennial Central Bank Survey.

- The impact of the LIBOR transition has been pronounced and has contributed to a virtual cessation of trading of FRAs that reference Libor.
- The daily turnover of FRAs had contributed \$1.9 trillion, or 30%, to the global total in the 2019 Survey, but only \$0.5 trillion (10%) in the 2022 Survey. FRAs denominated in US dollars led this decline

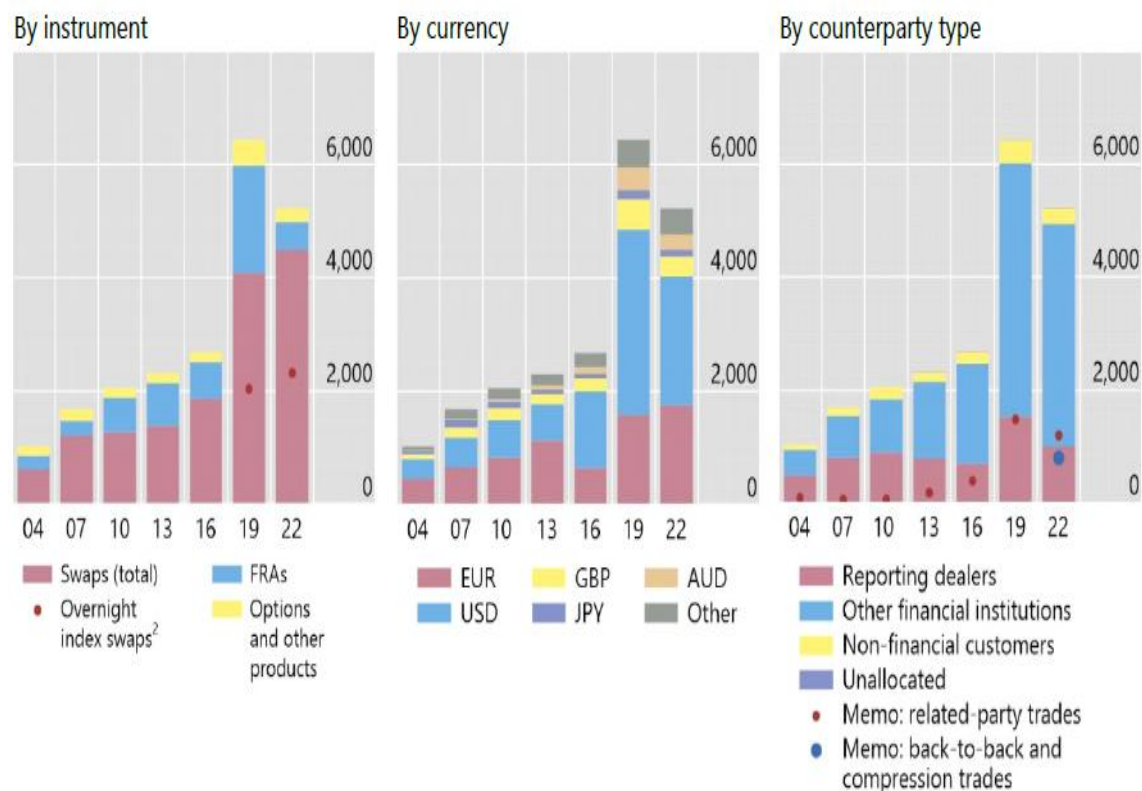
IRS dominates Interest Rate Derivative Markets; with USD & Euro deals dominating the market and struck mostly by OFIs

79

Turnover of OTC interest rate derivatives

Net-net basis,¹ daily averages in April, in billions of US dollars

Graph 1



¹ Adjusted for local and cross-border inter-dealer double-counting. ² Overnight index swaps are included in total swap turnover. Data available only from 2019.

- Turnover of OTC interest rate derivatives averaged US\$5.2 tn per day (“net-net basis) in April 2022 [\$6.4 trillion in 2019]). This was led by FRA turnover falling 74% between surveys, from \$1.9 trillion (30% of the global total) to \$0.5 trillion (10%). The turnover of interest rate swaps grew by 10% to \$4.5 trillion.
- Turnover of US dollar contracts amounted to \$2.3 trillion in April 2022, or 44% of global turnover. This share is down noticeably from 2019 and 2016, when dollar contracts accounted for roughly half of global turnover, and reflected the disproportionate impact of the LIBOR reform on USD FRAs (turnover fell by 98%). The dollar’s share in turnover of instruments other than FRAs (ie swaps, options and other products) rose between 2019 and 2022.
- Turnover of euro contracts reached \$1.8 trillion in April 2022, or 34% of global turnover (from 25% in 2019). Turnover in EUR swaps was \$1.3 trillion, up 38% since 2019. Similarly, turnover of EUR
- Sales desks in the UK recorded the highest turnover, at \$2.6 trillion (“net-gross” basis) or 46% of global turnover (down from 51% in 2019).

FRAs shrinking in interest rate derivative instruments

80

OTC interest rate derivatives turnover by instrument¹

"Net-net" basis,² daily averages in April in billions of US dollars

Table 1

Instrument	2010	2013	2016	2019*	2022
Interest rate instruments	2,054	2,311	2,677	6,439	5,226
FRAs	600	749	653	1,902	496
Swaps	1,272	1,388	1,859	4,080	4,491
Overnight index swaps	...	...	...	2,036	2,317
Other swaps	...	...	...	2,044	2,174
Options and other products ³	182	174	166	456	238
<i>Memo:</i>					
Turnover at April 2022 exchange rates ⁴	1,789	2,001	2,596	6,367	5,226
Exchange-traded derivatives ⁵	7,693	4,698	5,066	7,752	8,523

¹ Single currency interest rate contracts only. ² Adjusted for local and cross-border inter-dealer double-counting (ie "net-net" basis). ³ The category "other interest rate products" covers highly leveraged transactions and/or trades whose notional amount is variable and where a decomposition into individual plain vanilla components was impractical or impossible. ⁴ Non-US dollar legs of foreign currency transactions were converted into original currency amounts at average exchange rates for April of each survey year and then reconverted into US dollar amounts at average April 2022 exchange rates. ⁵ Sources: Euromoney Tradedata; Futures Industry Association; The Options Clearing Corporation; BIS derivatives statistics. Foreign exchange futures and options traded worldwide. * Revised data.

- Turnover in single currency OTC interest rate derivatives averaged \$5.2 trillion per day in April 2022.
- This was 19% lower than in the April 2019 Survey (\$6.4 trillion per day), although the present survey took place during a period of changing expectations about the path of future interest rates in major currencies, lingering Covid-19 related disruptions, and rising geopolitical tensions following the Russian invasion of Ukraine & most importantly LIBOR transition related uncertainty.

What Do we Know about FX Markets beyond the 2022... (1)

FX Swaps Continue to Dominate FX Markets, followed by spot trades

81

OTC foreign exchange turnover by instrument

"Net-net" basis,¹ daily averages in April in billions of US dollars

Table 1

Instrument	2010	2013	2016	2019*	2022*	2025
Foreign exchange instruments	3,973	5,357	5,066	6,581	7,468	9,595
Spot transactions	1,489	2,047	1,652	1,979	2,085	2,957
Outright forwards	475	679	700	998	1,157	1,847
Foreign exchange swaps	1,759	2,240	2,378	3,198	3,798	3,986
Currency swaps	43	54	82	108	124	172
Options and other products ²	207	337	254	298	303	634
<i>Memo:</i>						
<i>Turnover at April 2025 exchange rates³</i>	3,506	4,648	4,751	6,320	7,311	9,595
<i>Exchange-traded derivatives⁴</i>	144	145	115	127	154	207

¹ Adjusted for local and cross-border inter-dealer double-counting (ie "net-net" basis). ² The category "other FX products" covers highly leveraged transactions and / or trades whose notional amount is variable and where a decomposition into individual plain vanilla components was impractical or impossible. ³ Non-US dollar legs of foreign currency transactions were converted into original currency amounts at average exchange rates for April of each survey year and then reconverted into US dollar amounts at average April 2025 exchange rates. ⁴ Sources: Euromoney Tradedata; Futures Industry Association; The Options

What Do we Know about FX Markets beyond the 2022... (2)

US Dollar continues to dominate currency market turnovers

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- The share of US dollar in FX currency markets that had increased from 87% in 2013 to 88.4% in 2022 has gone up further to 89.2% in 2025.
- The share of Euro that had fallen from 33.4% on 2013 to 30.6% in 2022, dropped steeply thereafter to 28.9% in 2025.
- The share of JPY that dropped markedly from 23.0% in 2013 to 16.7% in 2022. marginally edged up to 16.8% in 2025
- GBP share that had gone up from 11.8% in 2013 to 12.9% in 2022, fell markedly to 10.2% in 2025
- The share of CNY from 2.2% in 2013 to 7.0% in 2022, increased further to 8.5% in 2025
- The share of INR that increased from 1.0% in 2013 to 1.6% in 2022, increased farther to 1.9% in 2025
- 98.9% of the exchange traded FX derivatives were in Us dollar in 2025

Global FX Code

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- The FX Global Code is a comprehensive set of global principles for foreign exchange (FX) market participants that promotes integrity, effective functioning, and transparency in the wholesale FX market.
- The Code outlines good practices in areas like ethics, governance, risk management, execution, information sharing, confirmation and settlement of trade practices and compliance to foster a robust, fair, and liquid market. It is a reference for market participants to align their internal procedures with international best practices and is supplemented by a public register where institutions can post their commitment to the Code
- Key Aspect of Global FX Code:
 - **Collaboration:** It was created through a partnership between central banks and private sector market participants from around the world
 - **Purpose:** To enhance the integrity and effective operation of the wholesale foreign exchange market by establishing unified, global guidelines for best practices.
 - **Scope:** The Code addresses principles for ethics, transparency, governance, risk management, and compliance, including specific practices for execution, confirmations, prime brokerage, and electronic trading.
 - **Voluntary Nature:** The FX Global Code is not a regulatory document but a set of principles and good practices that market participants can choose to adhere to.
 - **Promotes Fair Competition:** Aims to create a level playing field where participants can compete confidently and effectively based on transparent

Evolution of the Global FX Code

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- The FX Global Code is a comprehensive set of global principles for foreign exchange (FX) market participants that promotes integrity, effective functioning, and transparency in the wholesale FX market.
- The Code was developed by a partnership between central banks and market participants from 20 jurisdictions through the **Global Foreign Exchange Committee (GFXC)** under the auspices of Bank for International Settlements (BIS). GFXC was established in May 2017 as a forum bringing together central banks and private sector participants with the aim to promote a robust, fair, liquid, open, and appropriately transparent FX market in which a diverse set of participants, supported by resilient infrastructure, are able to confidently and effectively transact at competitive prices that reflect available information and in a manner that conforms to acceptable standards of behaviour.
- Its first version released in **May 2017**. It was subsequently revised in Dec. 2017; August 2018; July 2021 with its fifth version released in **Dec. 2024** that enhanced the Disclosure Cover Sheets for Liquidity Providers.
- The Code has been operationalized through:
 - **Statements of Commitment:** Central banks maintain a public register where market participants can post their Statements of Commitment to the Code.
 - **Internal Procedures:** Market participants use the Code as a reference to develop and review their internal procedures for conducting business in the wholesale FX markets.
 - **Local Adoption:** Local FX Committees work to promote adherence to the Code and provide updates on its implementation in different markets
 - Global FX Code is backed by **proportionality and self-assessment tools** based on identified 55 Principles of the Code that determines the proportion in which the FX market participant is able to assess his expected commitments to the Code.

What is Bank for International Settlements (BIS)?

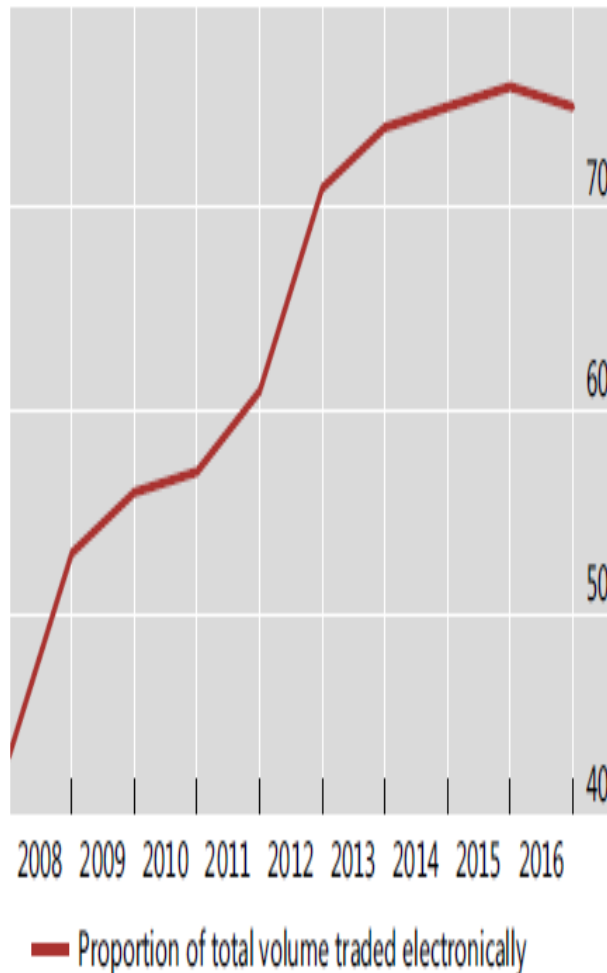
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- The BIS is an IFI owned by 63 central banks representing countries from around the world that together account for about 95% of world GDP. Its head office is in Basel, Switzerland & it has two representative offices in Hong Kong SAR & in Mexico City.
- It runs Innovation Hub, operational for last 5-years with 7 centres (in Frankfurt/Paris, Hong Kong SAR, Singapore, Switzerland, London, Stockholm and Toronto). These hubs have completed 31 projects completed and another 26 are under way.
- BIS runs some very important central bank committees to foster global financial cooperation and contribute towards maintaining systemic stability of the global financial system. These include
 - The **Basel Committee of Banking Supervision (BCBS)** for global standard setting for the prudential regulation of banks.
 - The **Committee on the Global Financial System (CGFS)** which is the central bank forum for the monitoring and dealing with issues relating to financial markets and suggesting policies for monetary and financial stability.
 - The **Markets Committee (MC)** for discussing market conditions & functioning & central bank operations. It is responsible for Triennial Surveys on FX & OTC Derivatives Markets & was instrumental in setting up GFXC for the Global FX Code.
 - The **Committee on Payments and Market Infrastructures (CPMI)** for promoting safety & efficiency of payment, clearing, settlement and related arrangements, thereby supporting financial stability and the wider economy
 - **Central Bank Governance Forum** to foster the good governance of central banks as public policy institutions.
 - **The Irving Fisher Committee on Central Bank Statistics (IFC)** for discussing statistical issues of interest to central banks.
- The BIS also hosts the secretariats of some other bodies like the **Financial Stability Board (FSB)**, the **International Association of Insurance Supervisors**, and the **International Association of Deposit Insurers (IADI)**
- BIS has also established **Financial Stability Institute (FSI)** for research and capacity building of supervisors.

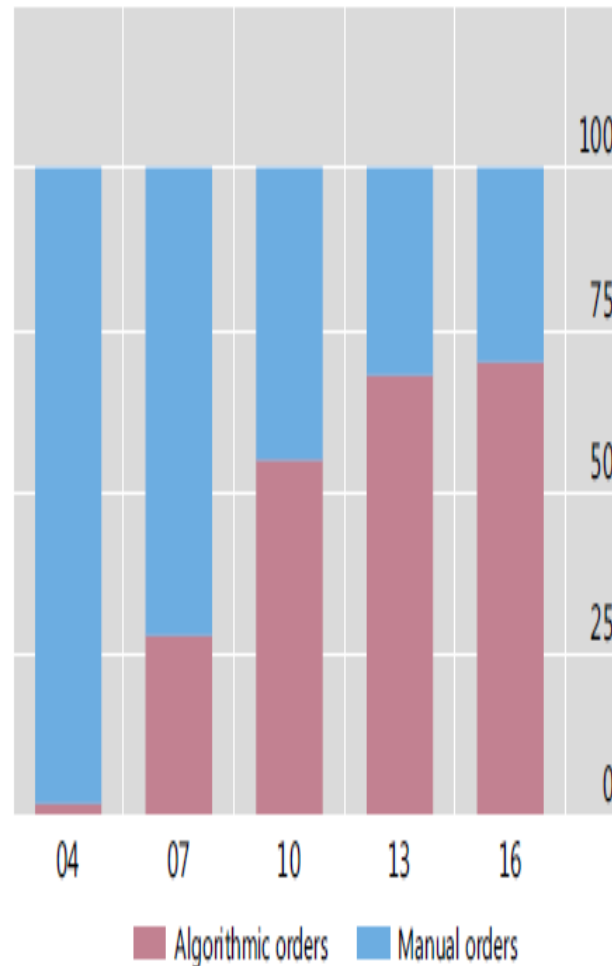
Rapidly Rising Share of algorithmic HFT trading in FX markets

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Shares of 2016 spot FX turnover executed electronically



Share of algorithmic activity in spot FX on EBS



- In the spot FX, end-users can now access liquidity via a range of electronic platforms & participate in increasingly automated market. Survey data suggest that, since 2013, over 70% of spot trading are executed electronically
- ~70% of orders on Electronic Broking Services (EBS) platform offered by the CME group, a primary central limit order book (CLOB) and a major inter-dealer platform for spot FX, are now submitted by algorithms, rather than manually.
- EBS's closest competitor is London Stock Exchange Group's FX Matching service (LSEG Matching).
- As part of the concerns with HFT FX trading, ParFX launched 'Randomization' on its platform, an application of a randomized pause of 20-80 milliseconds to eliminate speed advantage based on aggressive algos that benefit latency arbitrageurs. Several platforms like EBS, Thomson Reuters & LSEG followed. Some of the platforms, including the dominant EBS, have reversed some of these measures of late.

What Are Fast-paced Electronic Markets and why they need to be monitored?

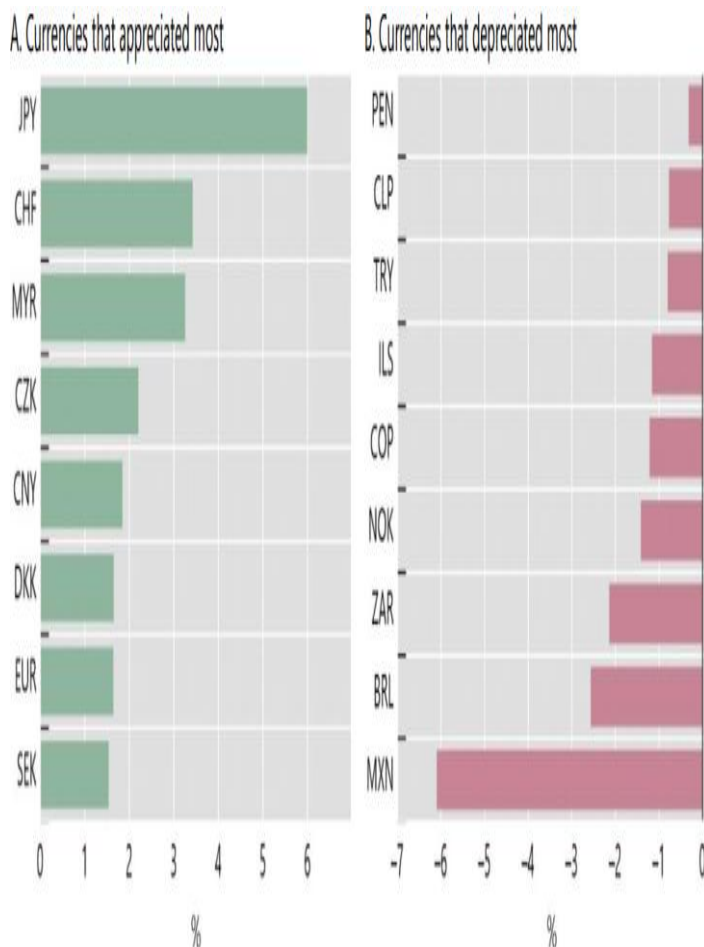
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- **BIS Project Rio (launched 12 Apr 2023)**
- Rio is a cloud-based live monitoring tool connecting to multiple FX trading venues and analysing activity as it occurs. By using data streaming technology as a real-time digital assembly line, market dislocations, liquidity issues and spillovers are identified as they occur. An intuitive and interactive dashboard designed with a group of central bankers visualises developments.
- Today's fast and interconnected markets generate millions of transactions every second. Central banks need to monitor and understand these markets to implement policy, manage reserves and assess risks. Yet making sense of the complexity of data created by fast-paced markets is a technology challenge.
- Rio meets this challenge. It is a live monitoring tool connecting to multiple FX trading venues and analysing activity as it occurs. By using data streaming technology as a real-time digital assembly line, market dislocations, liquidity issues and spillovers are identified as they occur. An intuitive and interactive dashboard designed with a group of central bankers allows visualising developments.
- Rio will evolve with global markets. Its underlying technologies are open source with a modular architecture that allows adding new markets, metrics and visuals simply and quickly. This includes use cases beyond financial markets. Rio is just the beginning for central banks.

What is Currency Carry Trade?

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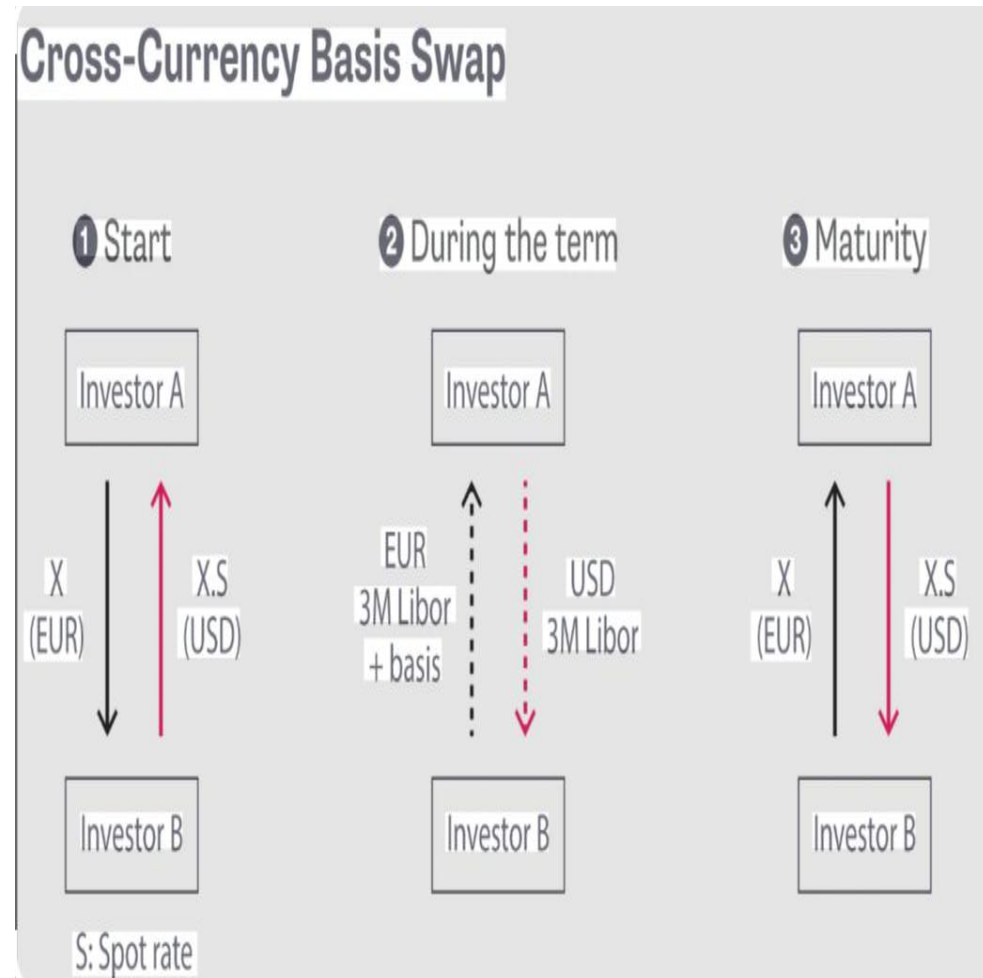
The Great Carry Trade Unwinding of 2024:
Changes in exchange rate vs. US dollar
during Aug 1-5, 2024



- A carry trade is an investment strategy where you borrow money at a low interest rate to invest in assets that offer a higher return. The profit, or "carry," is the net difference between the yield you earn on your investment and the cost of your borrowing.
- Financial market volatility surfaced in early August 2024 on account of the unwinding of leveraged trades in equity and currency markets. The leveraged positions amplified the market reaction to a negative macro release in the U.S. However, markets stabilized quickly and volatility receded. FX carry trades were hit hard by the deleveraging pressures. Their overall size is difficult to measure. Various estimates based on both on- and off-balance sheet activity yield a rough middle ballpark of ¥40 trillion (\$250 billion) going into the event, which, if anything, is biased down due to data gaps.
- The peak of the stress occurred on 5 August, when the Japanese TOPIX index lost 12% in a single day, and the VIX briefly registered levels not seen since Covid-19. The event was yet another example of volatility exacerbated by procyclical deleveraging and margin increases. Although an outright market dysfunction was averted this time, the structural features of the system underpinning such episodes deserve continued attention by policymakers.
- Popular currencies for funding carry trade are JPY, euro & CHF and investing in GBP, AUD, NZD. However, currency trades have also been used for exotic currencies. Exotic currencies are thinly traded currencies that suffer from illiquidity and have large bid-ask spreads. It includes BRL, IDR, INR, MYR, SGD, TRY & ZAR. Yen carry trades were used intensively to invest in INR assets. According to some currency strategists, INR was in October 2023 among Asia's top contenders for carry trades even though the currency was trading near its record low. Think why? RBI gave indications in last few monetary policies, it will hold high policy rates for long

Cross currency basis trade a significant FX arbitrage tool

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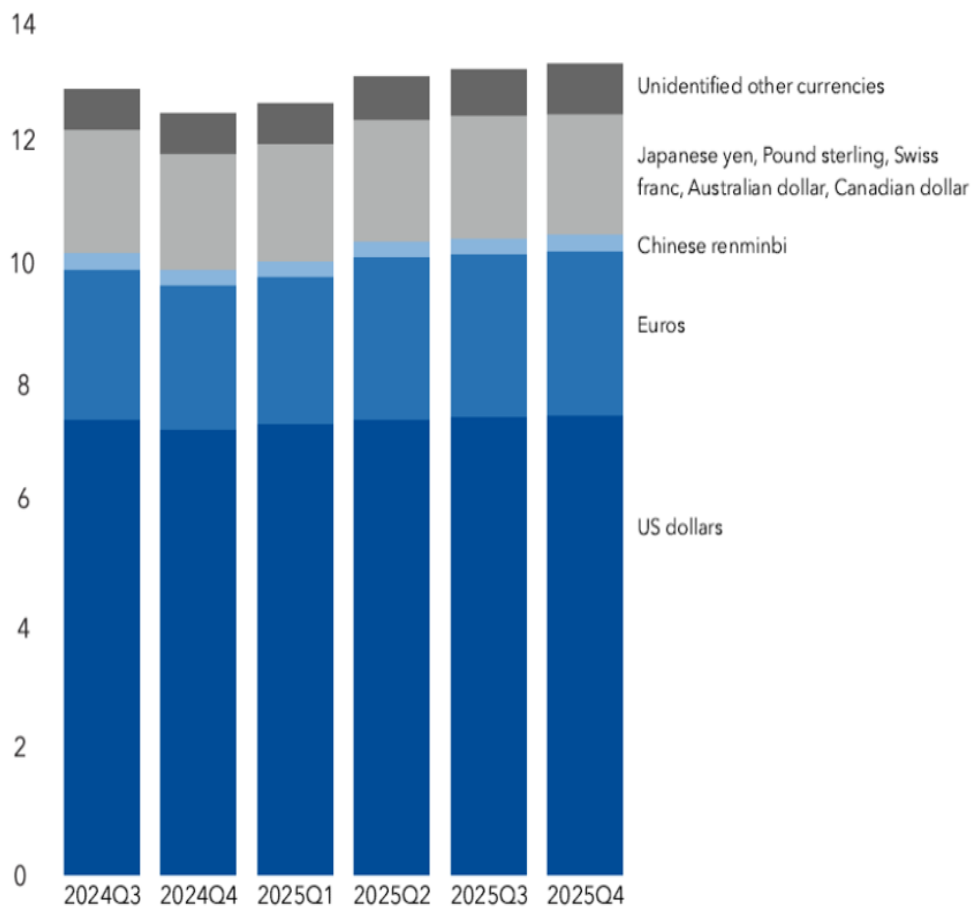
- **Cross-currency swaps** are commonly undertaken to arbitrage by exploiting the supply-and-demand mismatch between two currencies by simultaneously borrowing in one currency and lending in another via a cross-currency swap
- the **basis** is an additional spread added to or subtracted from one of the floating interest rate legs (usually the non-USD leg) to account for supply-and-demand imbalances, relative funding costs, and liquidity for a specific currency in global markets
- The basis exists and fluctuates based on several market factors. For example, (i) the US dollar is the most sought-after currency for international trade and reserves. A higher demand for dollar funding widens the basis; (ii) imbalances in international capital flows may shift desire to hedge the asset causing the basis to shift; (iii) Credit & Liquidity Risks, such as differences in creditworthiness and liquidity between two central banking systems can cause basis to shift.

De-dollarization is stark even if a multi-year process; Share of FX reserves held in US dollars has to 56.8% at end-2025 from ~70% at end-1999 (peak 72% in 2001)

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Reserves in the Fourth Quarter of 2025

Trillions of US dollars



- De-dollarization is occurring as a slow-burner process. US dollar share in allocated FX reserves of IMF member countries has dropped from ~70% at end-1999 (peak 72% in 2001) to 56.8% at end-2025 – an over 13 ppt drop in about a quarter century or ~15% in last 24 years.
- However, dollar may still remain as a reserve currency in the foreseeable future. Most trade is still invoiced in dollars and most financial transactions still get done in US dollars. In Europe ~66% of trade is invoiced in euros. Invoicing is predominantly in dollars elsewhere (96% in Americas, 74% in Asia-Pacific & 79% in ROW).
- As most trade transactions are dollar denominated, a currency depreciation does not help EMEs increase exports. Gita Gopinath explained this as a “**Dominant Currency Paradigm**”. It instead leads to costlier imports and high inflation.
- As of Q1 2025, \$9 trillion or 32% of marketable Treasury securities outstanding were held by foreign investors, both official and private. Over 40% of US dollar bank notes are held by foreigners. By April 2025, the total market capitalization of dollar stablecoins had reached about \$220 billion. If intra-euro area payments are included, the dollar share is ~60%

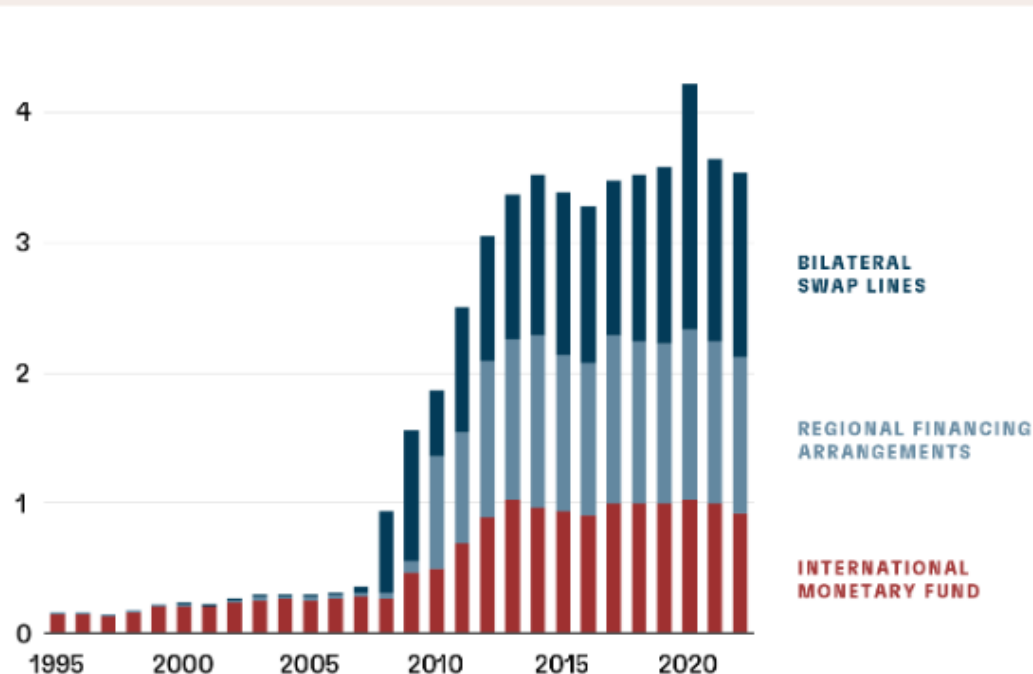
Swap Lines to deal with Dollar Shortages – bilateral swap lines & multilateral (Regional Financing Arrangements) providing GFSN

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International insurance mechanisms

Pooling reserve holdings between countries and drawing upon them when needed is more efficient; this is where the other components of the global financial safety net come in.

(trillions of US dollars)



SOURCES: Central bank websites; regional financing arrangements' annual reports; and IMF staff estimates.

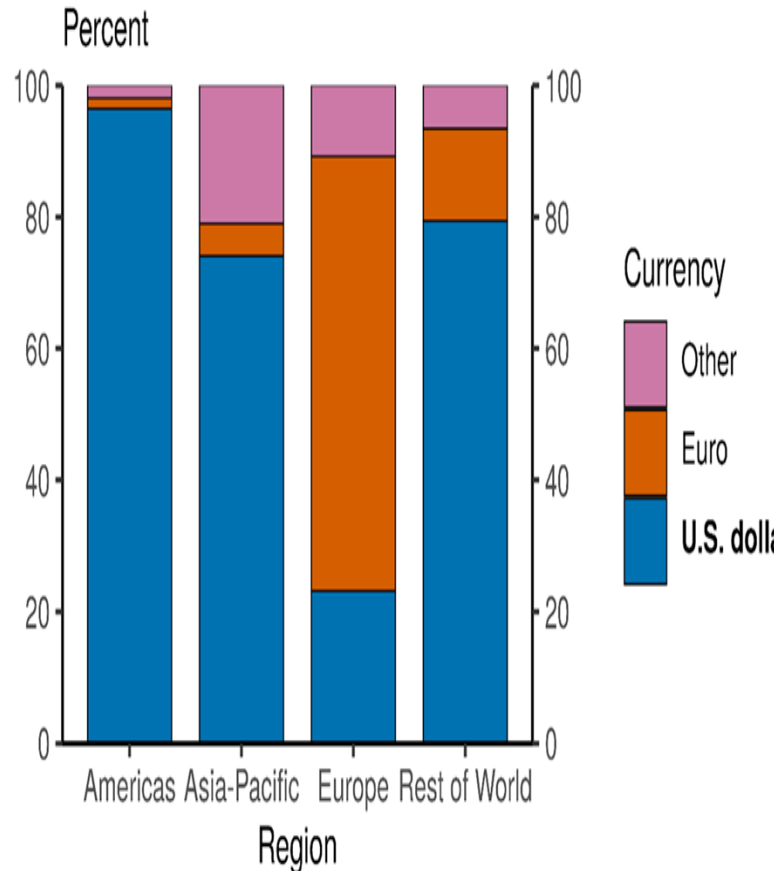
NOTE: Since the safety net is composed of various currencies, its US dollar value fluctuates with exchange rates.

- Bilateral swap lines such as US\$75 bn swap arrangement between India & Japan or
- the SAARC Currency Swap Framework of RBI to provide support to SAARC member countries (2024-27 INR Swap Window (₹25,000 crore) and a USD/Euro Swap Window (US\$2 billion).
- Regional Financing Arrangements like ESM, CMIM, BRICS CRA, FLAR aim at reserves pooling amongst central banks to strengthen Global Financial Safety Net (GFSN)
- Most RFAs have linked and delinked positions but it is important to stitch these layers of GFSN well

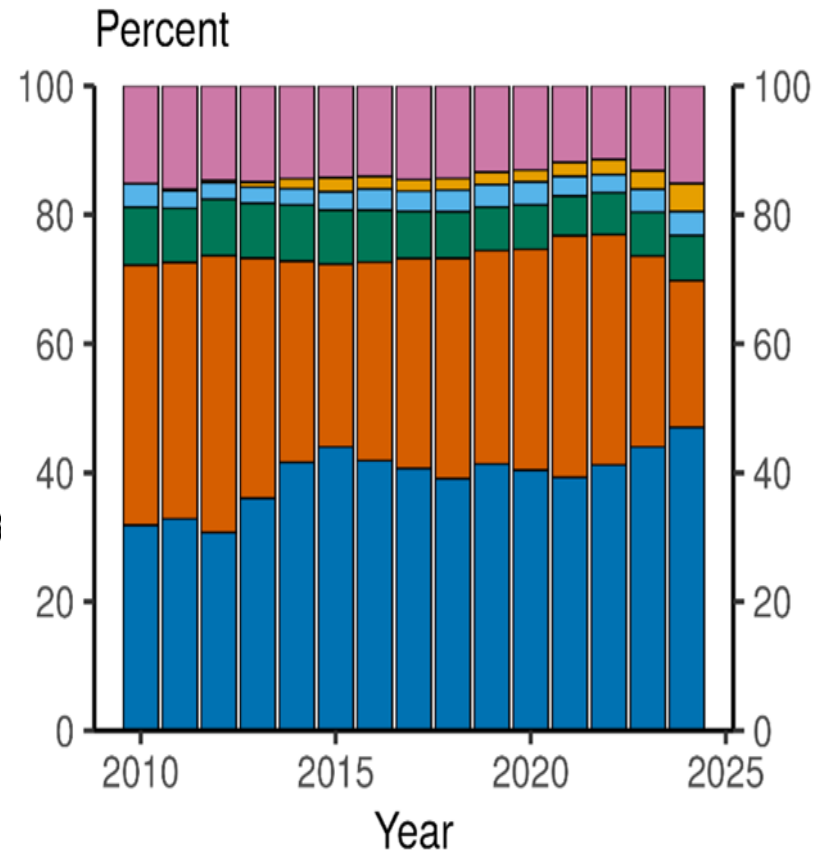
De-dollarization is a much weaker trend when it comes to Trade

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Share of US dollar & other currencies in Export Invoicing (%)



Share of US dollar & other currencies in International Payments (%)

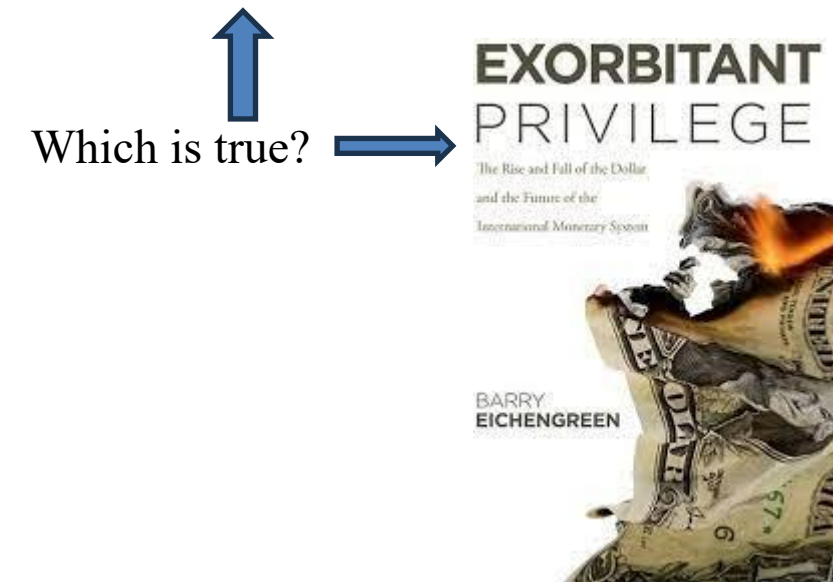


- Over the period 1999-2019, the dollar accounted for 96% of trade invoicing in the Americas, 74% percent in the Asia-Pacific region, and 79% in the rest of the world. The only exception is Europe, where the euro is dominant with 66%.
- The U.S. dollar share of international payments is about 50 percent and has even slightly increased in recent years. If intra-euro area payments are included, the dollar share is even higher at about 60 percent.

Should US take a larger burden share of ensuring a stable international trade & finance system given that US dollar has a reserve currency status?

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- **TRIFFIN DILEMMA/ PARADOX:** The Triffin dilemma or Triffin paradox is the conflict of economic interests that arises between short-term domestic and long-term international objectives for countries whose currencies serve as global reserve currencies. This dilemma was identified in the 1960s by Belgian-American economist, Robert Triffin, who pointed out that the country whose currency, being the global reserve currency, foreign nations wish to hold, must be willing to supply the world with an extra supply of its currency to fulfill world demand for these foreign exchange reserves by running a trade deficit.
- **EXORBITANT PRIVILEGE:** The term was originally coined by French finance minister, Valéry Giscard d'Estaing, in the 1960s to emphasize the unique role of the US dollar in the Bretton Woods system which allows the United States to lower external financing costs, since US debt provides international official investors with insurance against volatile capital flows and exchange rate risk. Demand by official investors for safe and liquid foreign currency assets to acquire and hold foreign exchange reserves reduces the financing costs of governments in reserve currency countries.



What are Special Drawing Rights (SDRs) & What role do they have in FX markets?

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Board-approved SDR Basket Currency Weights at Past Quinquennial Reviews

	U.S. dollar	Euro	Japanese yen	Pound sterling	Chinese renminbi	Sum
2000	44.20	29.40	14.80	11.60	-	100.00
2005	42.92	34.14	11.48	11.46	-	100.00
2010	41.94	37.36	11.26	9.44	-	100.00
2015	41.73	30.93	8.33	8.09	10.92	100.00
2022	43.38	29.31	7.59	7.44	12.28	100.00

- The IMF created the SDR as a supplementary international reserve asset in 1969, when currencies were tied to the price of gold and the US dollar was the leading international reserve asset. The IMF defined the SDR as equivalent to a fractional amount of gold that was equivalent to one US dollar.
- When fixed exchange rates ended in 1973, the IMF redefined the SDR as equivalent to the value of a basket of world currencies. The SDR itself is not a currency but an asset that holders can exchange for currency when needed. The SDR serves as the unit of account of the IMF and other international organizations.